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Influence of initial phase angle on optimally windowed strain-controlled chirp (γ – OWCh) rheometry

Mohua Das,[1,a)] Damien C. Vadillo,[2,b)] Alessandro Perego,[2,c)] and Gareth H. McKinley[1,d)]

[1]*Hatsopoulos Microfluids Laboratory, Department of Mechanical Engineering, Massachusetts Institute of Technology, Cambridge, Massachusetts 02139*
[2]*Corporate Research Analytical Laboratory, 3M, Maplewood, Minnesota 55144*

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Abstract

Optimally windowed chirp rheometry—a technique employing both frequency and amplitude-modulated exponential chirps—is gaining popularity for its ability to precisely capture the linear viscoelastic spectrum of complex materials while dramatically reducing the experimental time required to acquire data. To date, the chirp signals used for characterizing the linear viscoelasticity of such materials have featured a high time-bandwidth product (TB $>$ 50), calculated as the product of chirp duration or experiment time (T_{owc}) and frequency bandwidth ($\Delta\omega$). However, for time-evolving materials, it is essential to also adjust the chirp duration (T_{owc}) according to the characteristic mutation timescale, $\tau_{mu}(t)$ of the material to ensure time-translation invariance ($T_{owc} \ll \tau_{mu}(t)$) during experiments. For rapidly mutating materials, the values of the time-bandwidth product TB thus systematically decrease as the chirp duration is progressively reduced during the mutation process. However, the signal must also be sufficiently long, relative to the material's inherent relaxation timescale (τ), to ensure that a steady-state periodic viscoelastic response is obtained. To explore the trade-offs between these two competing conditions, we perform exploratory numerical simulations using canonical constitutive models: the classical Maxwell model, characterized by a single dominant relaxation time and the fractional Maxwell model, which compactly and accurately captures the broad relaxation spectra typical of many viscoelastic materials. For highly viscoelastic systems and/or rapidly mutating samples, we show that when employing short exponential chirp sequences, substantially improved accuracy can be obtained by shifting the initial phase of the chirp waveform through a single new phase offset parameter, φ_0. Experimental validation tests performed on wormlike micellar solutions, a pressure-sensitive adhesive, and a UV-curable acrylate system—each having relaxation times comparable to the signal duration—further demonstrate that adjusting the initial phase significantly enhances the data quality of linear viscoelastic measurements made with optimally windowed chirp protocols. © *2026 Author(s). All article content, except where otherwise noted, is licensed under a Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).* https://doi.org/10.1122/8.0001118

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I. INTRODUCTION

A key challenge in current rheological test protocols is accurate measurement of the rapidly evolving viscoelastic properties of "mutating" materials [1]. These are dynamically evolving systems that can undergo significant changes over the course of a single experiment due to changes in the molecular structure (such as polymerization, degradation, or cross-linking), phase transitions, or evolving microstructure [2].

In recent decades, chirp rheometry—a method that employs frequency-modulated deterministic signals—has become increasingly popular for investigating the linear viscoelastic behavior of a range of soft materials [3–11]. This technique offers several advantages over conventional discrete frequency sweep (DFS) experiments [12] as it significantly reduces experimental duration and avoids the large strain amplitudes often encountered in superposed multiwave techniques [2], which can exceed the linear viscoelastic regime. The low crest factor of a chirp, i.e., the ratio of peak to root mean square value of waveform amplitude, along with its high signal-to-noise ratio (SNR), makes it an excellent solution for mechanical spectroscopy. One recent advancement in chirp-based rheometry, introduced by Geri *et al.* [7], is the development of the strain-controlled optimally windowed chirp (γ-OWCh) signal. This method combines both frequency and amplitude modulation into a single strain-controlled input signal in which the frequency varies exponentially in time. By incorporating a cosine-tapered (Tukey) windowing function [7,8], the γ-OWCh waveform effectively minimizes spectral leakage [13], which is particularly important for short-duration signals. OWCh rheometry remarkably achieves data quality similar to conventional DFS experiments, while reducing measurement time by up to 2 orders of magnitude for comparable spectral content and increasing data density [7,14].

The newly modified governing equations of the input forcing function for the OWCh signal with a user-specified phase shift are given in the following expressions:

[a)]Electronic mail: mohuadas@mit.edu
[b)]Electronic mail: dvadillo@mmm.com
[c)]Electronic mail: aperego@mmm.com
[d)]Author to whom correspondence should be addressed; electronic mail: gareth@mit.edu

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$$x(t) = x_0 \begin{cases} \cos^2\left[\frac{\pi}{r}\left(\frac{t-t_w}{T_{\text{owc}}} - \frac{r}{2}\right)\right] \sin\left[\frac{\omega_1 T_{\text{owc}}}{\log(\omega_2/\omega_1)}\left(\exp\left(\log(\omega_2/\omega_1)\frac{t-t_w}{T_{\text{owc}}}\right) - 1\right) + \varphi_0\right], & \frac{t-t_w}{T_{\text{owc}}} \le \frac{r}{2}, \\ \sin\left[\frac{\omega_1 T_{\text{owc}}}{\log(\omega_2/\omega_1)}\left(\exp\left(\log(\omega_2/\omega_1)\frac{t-t_w}{T_{\text{owc}}}\right) - 1\right) + \varphi_0\right], & \frac{r}{2} < \frac{t-t_w}{T_{\text{owc}}} < 1 - \frac{r}{2}, \\ \cos^2\left[\frac{\pi}{r}\left(\frac{t-t_w}{T_{\text{owc}}} - 1 + \frac{r}{2}\right)\right] \sin\left[\frac{\omega_1 T_{\text{owc}}}{\log(\omega_2/\omega_1)}\left(\exp\left(\log(\omega_2/\omega_1)\frac{t-t_w}{T_{\text{owc}}}\right) - 1\right) + \varphi_0\right], & \frac{t-t_w}{T_{\text{owc}}} \ge 1 - \frac{r}{2}, \end{cases} \quad (1)$$

where ω_1 and ω_2 represent the initial and final oscillatory test frequency of interest, φ_0 is the user-specified phase shift, and t_w is an initial waiting time (typically $\sim O(1\,s)$) at the start of the chirp signal to ensure that the sensitive torque transducer has settled at a well-defined baseline. The instantaneous frequency $\omega(t)$ evolves exponentially over the chirp duration T_{owc} and is given by [7]

$$\omega(t - t_w) = \omega_1 \left(\frac{\omega_2}{\omega_1}\right)^{(t-t_w)/T_{\text{owc}}}. \quad (2)$$

For the γ-OWCh signal, $x(t)$ in Eq. (1) represents the amplitude of the time-varying strain, $\gamma(t)$, and x_0 represents the maximum strain amplitude, γ_0, which must be chosen within the linear viscoelastic limit. The windowing parameter, r, ensures periodicity of the input signal ($\gamma(t - t_w = 0) = 0$, and $\gamma(t = T_{\text{owc}} + t_w) = 0$). Under linear viscoelastic conditions, this tapering likewise makes the stress periodic, thereby reducing spectral leakage in both the input strain and the output stress spectra. According to Geri *et al.* [7], the optimal range for r in typical rheometric experiments is $0.06 \le r \le 0.15$. The chirp duration is commonly related to the initial frequency via $T_{\text{owc}} \ge (2/3) \cdot (2\pi/\omega_1)$ [7,14]. In all existing rheometric applications of the optimally windowed chirps, the initial phase shift of the chirp signal has been set to zero, $\varphi_0 = 0°$. The material's complex modulus $G^*(\omega)$ is computed in the frequency domain using the following expression:

$$G^*(\omega) \equiv \frac{\text{FFT}\{\sigma(t)\}}{\text{FFT}\{\gamma(t)\}} = |G^*(\omega)|e^{i\delta(\omega)} = G'(\omega) + iG''(\omega), \quad (3)$$

where $G'(\omega) = |G^*(\omega)|\cos(\delta(\omega))$ is the storage (or elastic) modulus, $G''(\omega) = |G^*(\omega)|\sin(\delta(\omega))$ is the loss (or viscous) modulus, $\delta(\omega)$ is the phase angle, $\tan(\delta(\omega)) = G''(\omega)/G'(\omega)$, and FFT denotes the Fast Fourier Transform [16].

The γ-OWCh protocol can be implemented on Separate Motor Transducer (SMT) rheometers that operate in a strain-controlled mode, such as the TA Instruments ARES-G2 (as described by Geri *et al.* [7]) and the Anton Paar MCR702 TwinDrive, as demonstrated by Athanasiou *et al.* [17]. More recently, Hudson-Kershaw *et al.* [15] extended the applicability of optimally windowed chirp protocols to Combined Motor Transducer (CMT) systems by introducing the stress-controlled OWCh (σ-OWCh) protocol, in which now the shear stress, $\sigma(t)$, is the input forcing function $x(t)$ and the strain, $\gamma(t)$, and strain rate, $\dot{\gamma}(t)$, are the output responses. To mitigate spectral leakage caused by finite-time strain offsets in the σ-OWCh method, the following expression is used to compute the material's complex modulus:

$$G^*(\omega) = i\,\omega\,\eta^*(\omega), \quad \text{where} \quad \eta^*(\omega) = \frac{\text{FFT}\{\sigma(t)\}}{\text{FFT}\{\dot{\gamma}(t)\}}, \quad (4)$$

$\eta^*(\omega)$ represents the complex viscosity of the material. This development significantly broadens the utility of OWCh rheometry, enabling its application across a wide range of rheometric platforms.

Although OWCh rheometry is applicable to a broad range of materials, its greatest utility lies in characterizing the rapidly evolving viscoelastic spectra of mutating material systems. As Athanasiou *et al.* [17] highlight, accurately capturing the linear viscoelastic (LVE) response of time-evolving materials is analogous to high-speed photography: just as the exposure time must be short to avoid motion blur, rheological measurements must be completed within a time window that is short relative to the *mutation* time [1] of the material. The mutation time, $\tau_{\text{mu}}(t)$, of the material is defined as follows [1]:

$$\tau_{mu}(t) = \left(\frac{\partial \ln g}{\partial t}\right)^{-1}_{\omega}, \quad g = \text{property of interest} = G^*(\omega, t);\ G'(\omega, t);\ G''(\omega, t). \quad (5)$$

To ensure that sample mutation is negligible during the experiment, the dimensionless mutation number (Mu) [1,18], defined as the ratio of the experimental time, $T_{\text{exp}}(= T_{\text{owc}})$, to the mutation time, $\tau_{\text{mu}}(t)$, must satisfy the following condition:

$$Mu = \frac{T_{\text{owc}}}{\tau_{\text{mu}}(t)} \ll 1. \quad (6)$$

For gelation experiments specifically, values of $Mu < 0.15$ are recommended to ensure sufficiently time-resolved data during measurement [1].

However, satisfying this constraint on the mutation number is only part of the challenge in time-resolved rheometry. Any such time-resolved measurement must take into account three crucial timescales [1]: the material's mutation time ($\tau_{\text{mu}}(t)$), the inherent dominant relaxation time of the material (τ), and the experimental duration—in our case, T_{owc}. While T_{owc} must be short enough to ensure that the material does not mutate appreciably over the measurement window (i.e., $Mu \ll 1$), it must also be sufficiently long to allow sufficient time to ensure that the initial transient

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response of a viscoelastic material decays to zero within the duration of the OWCh experiment: a process governed by the material's dominant relaxation time (τ). The ratio τ/T_{owc} defines the Deborah number (De) [19–21] of the rheological test, which must remain much less than one to ensure a steady-state periodic viscoelastic response. Notably, in an OWCh experiment, the frequency, $\omega(t)$ [Eq. (2)], evolves continuously over the duration of the test, making the protocol inherently unsteady. However, when defining the Deborah number (De), we focus primarily on the initial frequency (ω_1) of the signal since this initial value also determines the total chirp duration (T_{owc}). For the rapidly oscillating OWCh signals, by a "steady state," we mean the periodic harmonic response (also referred to as the alternance state [22]) that is attained after the secular terms associated with the initial viscoelastic transient response of the system have sufficiently decayed away. Figure 1 illustrates the design space for mutating materials, with the horizontal axis representing the Deborah number (De) and the vertical axis indicating the mutation number (Mu), with maximum thresholds of $De_{\max} = 1$ and $Mu_{\max} = 0.15$.

However, as highlighted by Holly *et al.* [2], rheological experiments face a trade-off between two competing time-dependent mechanisms: viscoelastic stress relaxation toward the time-periodic alternating steady state and sample mutation. The nondimensional constraint that satisfies both stability criteria simultaneously is [1,2]

$$\frac{\tau}{\tau_{\mathrm{mu}}(t)} \equiv \frac{\tau}{\left(\frac{\partial \ln g}{\partial t}\right)_{\omega}^{-1}} \ll 1. \tag{7}$$

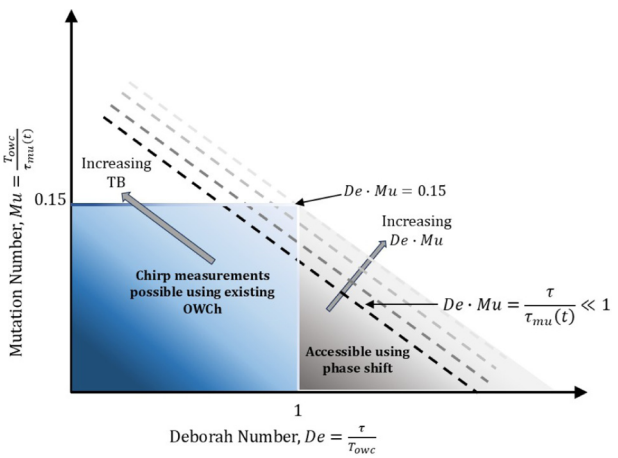


FIG. 1. Feasibility region for chirp-based rheological testing of mutating materials. The shaded (blue) region on the left represents the parameter space in which all constraints $De \ll 1$, $Mu \ll 1$, and $De \cdot Mu \ll 1$ are simultaneously satisfied, enabling accurate measurement of the viscoelastic properties using standard OWCh protocols. The gray arrow (left) shows how the operating point shifts as the time-bandwidth product $\mathrm{TB} = T_{\mathrm{owc}}(\omega_2 - \omega_1)/2\pi$ increases, i.e., when the chirp duration is increased to improve spectral resolution. While De decreases and Mu increases along this path, the test may eventually violate the constraint on the mutation number. The gray shaded region (right) illustrates the extended operating window accessible through the introduction of an optimized initial phase angle in the chirp signal, which allows reliable measurements even when the condition $De \ll 1$ is not strictly met.

We can rewrite the equation as follows:

$$\left(\frac{\tau}{T_{\mathrm{owc}}}\right) \cdot \frac{T_{\mathrm{owc}}}{\left(\frac{\partial \ln g}{\partial t}\right)_{\omega}^{-1}} \ll 1. \tag{8}$$

In Eq. (8), the first ratio is the Deborah number (De) and the second is the mutation number (Mu). Thus, the feasibility of accurate rheological testing on mutating materials relies on simultaneously fulfilling the conditions $Mu < 0.15$, $De \ll 1$ and the following combined dimensionless criterion:

$$De \cdot Mu \ll 1. \tag{9}$$

This combined criterion also serves as an indicator of a quasi-steady state of stress within the sample [1]. The blue-shaded region in the log-log representation of Fig. 1 illustrates the parameter space where all these conditions are satisfied.

However, choosing a chirp duration T_{owc} such that it satisfies both negligible material mutation ($Mu \ll 1$) and adequate stress relaxation toward a periodic steady state ($De \ll 1$) poses a practical challenge. In many cases, the dominant relaxation timescale may be unknown prior to linear viscoelastic testing, and it may also be evolving with time (e.g., in gelation processes [23]). Thus, the constraints $\tau \ll T_{\mathrm{owc}}$ and $T_{\mathrm{owc}} \ll \tau_{\mathrm{mu}}(t)$ cannot always be satisfied for all materials. Moreover, even if the mutation timescale permits rapid measurements, the short duration of the experiment can limit spectral resolution. This depends on the time-bandwidth product of the signal (TB) [24], which is defined as the product of the chirp duration (T_{owc}) and the spectral width ($\Delta\omega = \omega_2 - \omega_1$) under consideration,

$$\mathrm{TB} = T_{\mathrm{owc}} \cdot \left(\frac{\omega_2 - \omega_1}{2\pi}\right). \tag{10}$$

To date, experimental investigations using optimally windowed exponential chirps have typically employed signals with relatively large time-bandwidth products ($\mathrm{TB} > 50$). However, it remains unclear whether satisfying the constraint on TB and Mu alone suffices to guarantee high-quality spectra. Therefore, a detailed investigation into chirp signals with a short time-bandwidth product ($\mathrm{TB} \leq 50$) [25] is essential to expand the practical applicability of chirp rheometry. Additionally, further exploration is needed into how best to design rheological test protocols when the material's dominant relaxation time, τ, becomes comparable to, or even larger than, the chirp duration, T_{owc} (as represented by the gray shaded region in Fig. 1). Specifically, the following scenarios need to be examined:

(i) Conditions $\tau \ll T_{\mathrm{owc}} \ll \tau_{\mathrm{mu}}(t)$ is met, but the criterion $\mathrm{TB} > 50$ is not satisfied.
(ii) The time-bandwidth product (TB) is sufficiently high, but the condition $\tau \ll T_{\mathrm{owc}} \ll \tau_{\mathrm{mu}}(t)$ is not met.
(iii) Neither of the above conditions is satisfied.

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To explore these questions, we first conducted exploratory numerical simulations using canonical viscoelastic models, beginning with the classical Maxwell model (CMM) characterized by a single dominant relaxation time. In our simulations, we employed γ-OWCh signals with a large time-bandwidth product (TB $\approx$ 100); however, we found that when the chirp duration was comparable to or shorter than the material's relaxation time (i.e., $\tau/T_{\mathrm{owc}} \geq 1$), the stress response in the time domain did not preserve periodicity and symmetry throughout the duration of the chirp. Upon Fourier transformation, this in turn led to a noisy viscoelastic spectrum in the frequency domain. This highlights that a large TB product alone does not guarantee a symmetric and periodic response; the chirp duration (T_{owc}) must still need to exceed the characteristic relaxation time (τ) of the material to obtain reliable viscoelastic properties from OWCh experiments. A similar effect was observed for short-time-bandwidth signals (e.g., TB < 10), under the same condition $\tau/T_{\mathrm{owc}} \geq 1$.

Notably, satisfying the constraint $\tau/T_{\mathrm{owc}} \ll 1$ becomes even more challenging as TB decreases, since a lower time-bandwidth product inherently requires shorter T_{owc} (given that the final frequency of interest is limited by the instrument). From our initial numerical investigation, we conclude that even if T_{owc} is chosen to satisfy the constraint of a low mutation number [Eq. (6)], a significant residual viscoelastic stress remains in the deformed test sample at the end of the chirp whenever $\tau/T_{\mathrm{owc}} \not\ll 1$, regardless of the TB product. This incomplete transient viscoelastic relaxation process in the time domain contributes to spectral leakage in the frequency domain. Consequently, since the material functions $G'(\omega)$, $G''(\omega)$ are derived from ratios of the transforms of the stress and strain signals in the frequency domain, the resulting viscoelastic spectra become corrupted and noisy.

As noted earlier, the recent work of Hudson-Kershaw *et al.* [15] reported a similar issue (finite-time strain offsets) for stress-controlled OWCh (σ-OWCh) rheometry. Their proposed solution is to perform the Fourier analysis on the time derivative of the strain signal and first obtain the complex viscosity, $\eta^*(\omega)$, before converting it to the complex modulus using the relation $G^*(\omega) = i\omega\eta^*(\omega)$. By analogy, one could attempt a similar approach for strain-controlled OWCh (γ-OWCh) by differentiating the measured stress signal and computing the following transfer function [22]:

$$\kappa^*(\omega) = i\omega G^*(\omega) = \frac{\mathrm{FFT}\{\dot{\sigma}(t)\}}{\mathrm{FFT}\{\gamma(t)\}}. \tag{11}$$

However, this approach encounters a key limitation for strain-controlled chirps. In the high-frequency region, windowing reduces the stress amplitude, causing the instrument's noise floor to dominate. Numerical differentiation (required to evaluate the *stress rate* $\dot{\sigma}(t)$) would further amplify this noise, making such an approach challenging.

A second possibility is to follow the strategy of Sheridan *et al.* [11] and introduce a post-chirp interval to allow the stress to decay back to zero. This, however, would require an additional arbitrary waveform segment. The ARES-G2 Arbitrary Wave software module currently allows a maximum of four segments, making the addition of a fifth post-chirp segment infeasible with our present configuration. Moreover, while such a modification would help enforce periodicity, it would not fully resolve issues related to the lack of $\pm$ symmetry in the measured stress signal.

Thus, to address these limitations, in the present work we introduce an additional degree of freedom (or a tuning parameter) into the OWCh formulation of Geri *et al.* [7]: a finite initial phase angle (φ_0). As we show in detail in Secs. II–V, our investigations demonstrate that carefully tuning this initial phase can significantly reduce residual stress at the end of the chirp signal by realigning the positions of local crests, troughs, and zero crossings in the input and output waveforms. As indicated schematically by the gray shaded region in Fig. 1, this strategy effectively recovers additional parameter space where reliable chirp-based measurements become possible even when the constraint $De(=\tau/T_{\mathrm{owc}}) \ll 1$ is not satisfied. Although this approach does not eliminate the transient viscoelastic response, it substantially mitigates its impact, thereby improving signal periodicity, symmetry, and spectral quality. We further illustrate the benefit of this optimal initial phase angle for accurately capturing viscoelastic spectra under conditions where $\tau/T_{\mathrm{owc}} \not\ll 1$, using a fractional Maxwell model (FMM) that more realistically represents the broad relaxation spectra, characteristic of many viscoelastic materials. Moreover, experimental investigations on various complex viscoelastic material systems—including wormlike micellar solutions, high-strength acrylic adhesive 300 LSE (3M™), and UV-curable acrylate systems (3M™)—confirm the practical significance and general applicability of using an optimal initial phase angle (φ_0) to improve chirp-based rheological measurements.

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II. NUMERICAL INVESTIGATION

A. Single-mode Maxwell model

While time-resolved rheometry is governed by the interplay of three fundamental timescales—the material's mutation time ($\tau_{\mathrm{mu}}(t)$), its inherent relaxation time (τ), and the experimental duration (T_{owc}), we begin our numerical analysis by focusing on the interaction between just two time scales: τ and T_{owc}. Our primary objective is to examine how the material's relaxation timescale (τ) influences the accurate determination of rheological properties in the Fourier domain. Specifically, we aim to explore the extent to which the experimental duration (T_{owc}) can be shortened, relative to the material's dominant relaxation timescale (τ), without compromising the accuracy of the complex modulus data obtained from the exponential chirp used to probe the material response.

We first consider the canonical two-parameter CMM with a single relaxation timescale for our analysis. The Maxwell element is conceptually represented as a series arrangement of a linear viscous (Newtonian) dashpot and a linear elastic

(Hookean) spring. The one-dimensional constitutive equation for the single-mode CMM can be represented in the following form [26,27]:

$$\sigma(t) + \frac{\eta_0}{G_0}\frac{d\sigma(t)}{dt} = \eta_0\frac{d\gamma(t)}{dt}. \tag{12}$$

Here, G_0 represents the elastic modulus of the material and η_0 represents the constant Newtonian viscosity. The single characteristic relaxation time associated with the two mechanical elements is $\tau = \eta_0/G_0$.

For a single-tone sinusoidal strain input, $\gamma(t) = \gamma_0\sin(\omega t)$, the corresponding strain rate is

$$\dot{\gamma}(t) = \frac{\mathrm{d}\gamma(t)}{\mathrm{dt}} = \gamma_0\omega\cos(\omega t). \tag{13}$$

Substituting $\dot{\gamma}(t)$ into Eq. (12), the constitutive differential equation can be rearranged and expressed in the following integral form using an integrating factor $e^{t/\tau}$:

$$\sigma(t)e^{t/\tau} = \frac{\gamma_0\eta_0\omega}{\tau}\int_0^t e^{t'/\tau}\cos(\omega t')\,dt'. \tag{14}$$

By solving Eq. (14), we obtain the following expression for the time-varying shear stress, $\sigma(t)$ [28]:

$$\sigma(t) = \gamma_0\left[\underbrace{G_0\frac{(\tau\omega)^2}{1+(\tau\omega)^2}}_{G'(\omega)}\sin(\omega t) + \underbrace{G_0\frac{\tau\omega}{1+(\tau\omega)^2}}_{G''(\omega)}\cos(\omega t)\right] - \gamma_0 G_0\frac{\tau\omega}{1+(\tau\omega)^2}\exp(-t/\tau). \tag{15}$$

The oscillatory stress, $\sigma(t)$, can be decomposed into an in-phase component and a quadrature, 90° out-of-phase component with respect to strain, $\gamma(t)$, providing the well-known expressions for the storage modulus $G'(\omega)$ and the loss modulus $G''(\omega)$ [28]. When multiplied together, the relaxation timescale (τ) and the frequency of interest (ω, in rad/s) express the ratio of the material's intrinsic relaxation time to the cycle duration, where $t_{\mathrm{cycle}} \propto 1/\omega$. This product can also be recognized as the Deborah number (De) for a discrete frequency test.

The final term on the right of Eq. (15) corresponds to the transient response, which fades away and becomes negligible once the stress response reaches a steady state [28]. This transient response of the stress signal is commonly referred to as the secular term,

$$\sigma_{\mathrm{transient}}(t) = \gamma_0 G_0\frac{\tau\omega}{1+(\tau\omega)^2}\exp(-t/\tau). \tag{16}$$

According to Eq. (16), the secular term of the stress response is influenced not only by the exponential decay factor, $\exp(-t/\tau)$, but also by the Deborah number (De) and can also be written in terms of $\exp(-\omega t/\omega\tau) = \exp(-\omega t/De)$. Two important asymptotic limits of Eq. (16) can be derived for very low and very high De values, as follows:

- $De \ll 1$: $\sigma_{\mathrm{transient}}(t) \approx \gamma_0 G_0(De)\exp(-\omega t/De)$.
- $De \gg 1$: $\sigma_{\mathrm{transient}}(t) \approx \gamma_0 G_0(De)^{-1}\exp(-\omega t/De)$.

At very low De, the secular term vanishes because the exponential tail, $\exp(-\omega t/De)$, decays rapidly; at very high De, it vanishes because the prefactor $1/De$ becomes very small. Thus, our analysis of the linear Maxwell model shows that for a single-mode sinusoidal input forcing function, the effect of the secular term on the linear viscoelastic measurements becomes most pronounced as the De number approaches unity or, equivalently, for high oscillatory test frequencies near the crossover frequency ($\omega_c = 1/\tau$).

To summarize, ensuring that the transient stress response has fully subsided requires that the experimental duration for each frequency response measurement must be sufficiently long, typically several times the material's relaxation timescale τ. This guarantees that viscoelastic measurements are performed while the stress response is in a periodic steady state. Notably, the experimental duration for the DFS test consists of the waiting period before data collection, t_{delay} (reported in terms of delay cycles, delay time, or both) as well as the cycle time, $t_{\mathrm{cycle}} = 2\pi/\omega$.

To explore the impact of waiting time on the acquisition of oscillatory data $(G'(\omega), G''(\omega))$ from discrete frequency sweep experiments, we examine one representative case at $De\ (= \omega\tau) = 5$. For the analysis, a single-tone sinusoidal strain input with an amplitude of $\gamma_0 = 1$ and a frequency of $\omega = 1$ rad/s is used, and the corresponding stress output is simulated using Eq. (12). The following values of the Maxwell model parameters are considered: $G_0 = 38$ Pa, $\tau = 5$ s. The complex modulus, $G^*(\omega)$, is computed from the simulated stress response using Eq. (3).

Figure 2(a) illustrates the stress output for varying numbers of delay cycles, ranging from 0 to 5. Delay cycles represent the number of periods (whole or fractional) of sinusoidal deformation applied at the specified amplitude and frequency before data collection begins. For example, a delay cycle of 0 indicates that the stress output used for Fourier transformation is collected immediately after applying the sinusoidal deformation, while a delay cycle of 1 means that stress data are collected after waiting for one complete cycle or an equivalent waiting time $t_w = 2\pi/\omega$. Different colors in the figure highlight the portions of the stress output utilized for the Fourier transformation.

Another effective representation of the transient and steady-state stress output is the 2D Lissajous–Bowditch plot, which cross-plots the output shear stress, $\sigma(t)$, as a function of the input strain, $\gamma(t)$ [Fig. 2(b)]. From this plot, it is clear that approximately three delay cycles (equivalent to a delay time of 18.85 s or nearly four times the relaxation time constant τ) are required for the output stress to reach the elliptical limit cycle corresponding to a steady-state response, represented by the solid line. Figure 2(c) displays the amplitude spectrum of the stress response as a function of the number of delay cycles. For a time-translational invariant system, in

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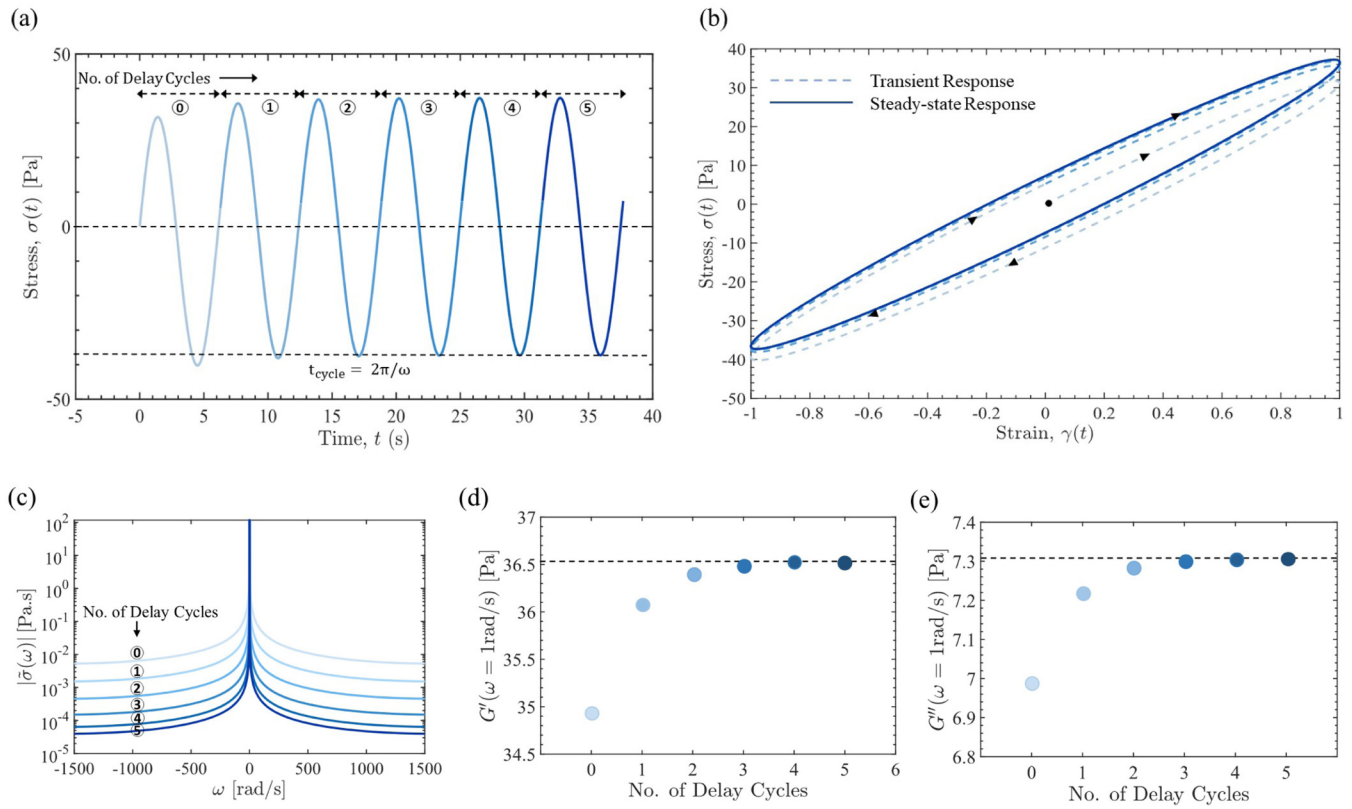


FIG. 2. Numerical simulation results using the single-mode CMM [Eq. (12)] with $\tau = 5$ s: (a) Stress response for a single-tone sinusoidal strain input, $\gamma(t) = \gamma_0 \sin(\omega t)$, with $\gamma_0 = 1$ and $\omega = 1$ rad/s. Different colors indicate portions of the stress signal used for the Fourier transform, with numbered symbols representing the corresponding number of delay cycles ($n\, t_{cycle} = n\, 2\pi/\omega$) considered. (b) 2D Lissajous–Bowditch curve of the stress response and input strain. Solid lines represent the steady-state response, while dashed lines indicate the transient response. At this test frequency (corresponding to $De = \tau\omega = 5$), the stress reaches a steady harmonic state after approximately three delay cycles. (c) Amplitude spectrum of the output stress for varying delay cycles or equivalent waiting times. (d) and (e) Computed values for the storage modulus, $G'(\omega = 1\,\text{rad/s})$, and loss modulus, $G''(\omega = 1\,\text{rad/s})$, as functions of the number of delay cycles or equivalent waiting time (t_w) considered before collecting data for the Fourier transform. Results converge after a waiting time (t_w) equivalent to three delay cycles.

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the limit when the experimental duration is $\gg \tau$, a perfect delta distribution, $\delta(\omega - 1)$, is expected at $\omega = 1\text{rad/s}$. Figures 2(d) and 2(e) show the computed values of $G'(\omega)$ and $G''(\omega)$ corresponding to the different number of delay cycles considered. The computed values of the storage modulus, $G'(\omega)$, and loss modulus, $G''(\omega)$, systematically converge as the number of delay cycles increases. These results highlight the importance of delay cycles (or selecting a finite waiting time during experiments) in order to ensure accurate and steady-state viscoelastic measurements.

During experiments, experienced rheologists typically estimate the relaxation timescale, τ, based on the material's behavior—whether it is more solid-like or liquid-like—while also considering the effects of temperature variations on material relaxation in time-temperature superposition tests. For materials with negligible rates of mutation, an appropriate waiting time, implemented as delay cycles, delay time, or a combination of both, is empirically selected to ensure accurate steady-state measurements. However, challenges arise when the time-translation invariance (TTI) [29] principle is not satisfied, i.e., when the material properties change considerably during the experimental window. In such cases, the experimental duration must be shortened. As discussed in Sec. I, this creates the dichotomous challenge of determining an experimental time that satisfies three dimensionless constraints simultaneously: $Mu \ll 1$, $De \ll 1$, and $De \cdot Mu \ll 1$.

As noted earlier, using OWCh signals with frequency and amplitude modulation has enabled precise determination of the time-evolving viscoelastic spectrum of mutating materials, while dramatically reducing the experimental duration [4,7,15,17]. With the ability to design chirp signals that satisfy the first dimensionless constraint, $Mu \ll 1$, a critical question arises: can the chirp duration (T_{owc}) be reduced to a value comparable to, or even lower than, the material's inherent relaxation time, τ? To the best of our knowledge, no closed-form solution exists for the output stress $\sigma(t)$ of Eq. (12) with an input γ-OWCh signal [Eq. (1)], owing to the mathematical complexity involved. Thus, numerical simulations using the canonical viscoelastic models were employed to investigate and quantify how far T_{owc} can be minimized.

For numerical analysis, we designed the input γ-OWCh signal [Eq. (1)] with an amplitude, $\gamma_0 = 1$, and initial frequency $\omega_1 = 0.3\,\text{rad/s}$, a final frequency of $\omega_2 = 30\,\text{rad/s}$, and a chirp duration $T_{owc} = 2\pi/\omega_1 = 20.9$ s. Thus, the time-bandwidth product of the resulting input γ-OWCh signal is $\text{TB} \approx 99$. A tapering parameter of $r = 10\%$ is applied, along with a waiting time of $t_w = 2$ s [$\gamma(t) = 0,\ \forall t \le t_w$] at the beginning of the chirp signal. All chirp-design parameters

are carefully selected to satisfy the criteria specified by Geri *et al.* [7] for a leakage-free spectrum in γ-OWCh rheometry.

We selected a sampling frequency, $f_s = 500\,\text{Hz}$, corresponding to a sampling interval $\Delta t = 0.002\,\text{s}$. This choice satisfies both the Nyquist–Shannon minimum sampling criterion [30] and the sampling requirement outlined by Hudson-Kershaw *et al.* [15] for chirp-based rheometry. Two sets of Maxwell model parameters are considered for the analysis. For the first set, we use $G_0 = 38\,\text{Pa}$ and $\tau = 1.19\,\text{s}$, representing the fitted parameters of a micellar solution modeled using the Maxwell model (Appendix C, Geri *et al.* [7]). For these parameters, the ratio $\tau/T_{\text{owc}} = 0.06$. For the second set, we retain $G_0 = 38\,\text{Pa}$ but consider a scenario where the sample's characteristic relaxation time, τ (=20.9 s) equals the chirp duration, T_{owc}, resulting in a ratio $\tau/T_{\text{owc}} = 1$.

The primary goal of selecting the above-mentioned chirp and Maxwell model parameters is to ensure that all conditions meet the optimal criteria for accurately determining the viscoelastic spectra $(G'(\omega), G''(\omega))$ through chirp experiments. By varying only one parameter—the material's inherent relaxation timescale (τ)—we aim to isolate and attribute any changes in the measured spectrum to the influence of the ratio τ/T_{owc}. From the numerical simulation, we expect the stress amplitude, σ_0, and the magnitude of the complex modulus $|G^*(\omega)|$ to satisfy the relationship $\sigma_0 = |G^*(\omega)|\gamma_0$, where

$$|G^*(\omega)| = \sqrt{G'(\omega)^2 + G''(\omega)^2}. \tag{17}$$

By substituting the expressions for $G'(\omega)$ and $G''(\omega)$ from Eq. (15), the expression for the complex modulus $|G^*(\omega)|$ simplifies to

$$|G^*(\omega)| = G_0\sqrt{\frac{(\tau\omega)^2}{1+(\tau\omega)^2}}. \tag{18}$$

The asymptotic behavior of $|G^*(\omega)|$ for very high and very low values of the Deborah number ($De = \tau\omega$) is as follows:

- $De \ll 1,\ |G^*(\omega)| \approx G_0\, De$
- $De \gg 1,\ |G^*(\omega)| \approx G_0\left(1 - \frac{1}{2De^2} + \cdots\right)$

Also, for the classical Maxwell model, the prediction of the phase angle is as follows:

$$\delta(\omega) = \tan^{-1}\left(\frac{G''(\omega)}{G'(\omega)}\right) = \tan^{-1}\left(\frac{1}{De}\right). \tag{19}$$

The corresponding plots for the input strain $\gamma(t)$, strain rate $\dot{\gamma}(t)$, output stress $\sigma(t)$, magnitude of the complex modulus $|G^*(\omega)|$, and phase angle $\delta(\omega)$ for the two cases described above are shown in Figs. 3(a)–3(e). Figure 3(a) illustrates the designed input strain chirp waveform (γ-OWCh), $\gamma(t)$ (black dashed line), along with its derivative, the strain rate $\dot{\gamma}(t)$ (black solid lines). The gray dotted envelope shows the Tukey window with a tapering parameter of $r = 10\%$, ensuring smooth transitions at the boundaries. Figure 3(b) displays the simulated output stress $\sigma(t)$ for the case of $\tau/T_{\text{owc}} = 0.06$. For this case, both the input strain and the output stress satisfy the periodicity condition, grow smoothly from, and return to, zero at $t = t_{\text{w}}$, and $t = T_{\text{owc}} + t_w$, respectively. Also, both signals exhibit symmetry about the centerline $(\gamma(t) = 0, \sigma(t) = 0)$ axis. The amplitude spectrum $|G^*(\omega)|$ and the phase angle $\delta(\omega)$ are computed from the ratio of the Fourier transforms of the output stress $\tilde{\sigma}(\omega)$ and input strain $\tilde{\gamma}(\omega)$, as defined in Eq. (3). For the case of $\tau/T_{\text{owc}} = 0.06 \ll 1$, both the amplitude spectrum $|G^*(\omega)|$ (shown as filled blue circles) and the phase angle $\delta(\omega)$ (shown as hollow blue rectangles) match the theoretical model predictions given by Eqs. (18) and (19). To quantify the agreement, the normalized mean absolute error (MAE) is presented in the inset of the same figure [Fig. 3(c)]. The MAE is calculated using the following equation:

$$\text{MAE},\ \varepsilon_A = \frac{1}{N}\sum_{i=1}^{N}\left|\frac{A_{\text{Simulated}} - A_{\text{CMM}}}{A_{\text{CMM}}}\right| \times 100\,\%, \tag{20}$$

where N is the total number of data points collected in the Fourier domain, and A represents the property of interest, namely, $|G^*(\omega)|$ or $\delta(\omega)$.

For the second case, where $\tau/T_{\text{owc}} = 1$, the output stress $\sigma(t)$ [Fig. 3(d)] exhibits deviations from the expected periodicity condition. This occurs despite the application of a windowed strain input during the simulation. The gray-filled symbols in Fig. 3(d) emphasize the deviation from symmetry in each cycle of the stress response [i.e., a cyclic stress offset changing with respect to time]. In the Fourier domain, lack of periodicity and symmetry of the output stress signal results in a noisy spectrum, particularly for the phase angle $\delta(\omega)$ [Fig. 3(e)]. The magnitude of the error observed for the phase angle is almost 118%, as indicated in the inset of Fig. 3(e). Thus, despite our input chirp signal adhering strictly to the optimal chirp-design criteria, notably a high time-bandwidth (TB) product and optimal Tukey window $r = 10\%$, we still obtain noisy spectra under conditions where $\tau/T_{\text{owc}} = 1$.

We analyze another extreme limit using the same Maxwell model parameters, $G_0 = 38\,\text{Pa}$ and $\tau = 1.19\,\text{s}$. This time, however, we consider the shortest possible time-bandwidth (TB) product. Specifically, we examine a frequency range spanning one decade, with $\omega_1 = 3\,\text{rad/s}$ and $\omega_2 = 30\,\text{rad/s}$, and the shortest possible chirp duration of $T_{\text{owc}} = 2\,\text{s}$ (TB ≈ 8.6). This results in a ratio of $\tau/T_{\text{owc}} = 0.6$. The corresponding curves are shown in Fig. 4. As we are considering the optimal amount of tapering, the aperiodic and asymmetric nature of the output stress, $\sigma(t)$ [Fig. 4(b)], can be attributed not only to the small value of TB but also to the comparable values of T_{owc} and τ. From Figs. 4(c) and 4(d), we can see significant impact of the noise in the computed values of the loss modulus, $G''(\omega)$ (red hollow circles), compared to the storage modulus, $G'(\omega)$ (red filled circles).

For the classical Maxwell model, at very high frequencies when $De \gg 1$,

- $G'(\omega) \approx G_0(1 - 1/De^2 + \cdots)$
- $G''(\omega) \approx \frac{G_0}{De} = G_0 \tan\delta(\omega)$ [$\because \delta(\omega) = \tan^{-1}(1/De)$]

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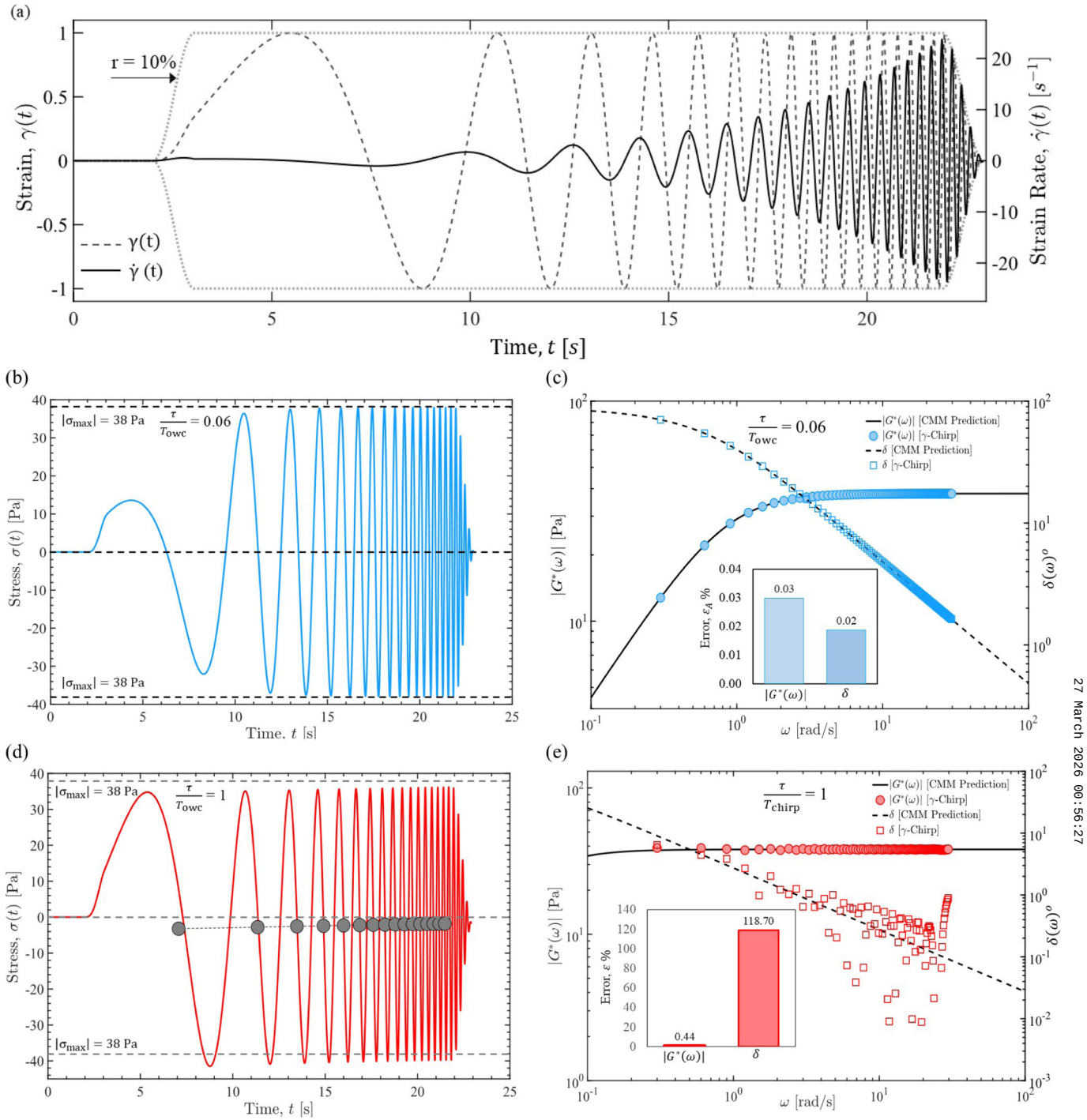


FIG. 3. (a) The input strain chirp waveform, γ-OWCh, is designed with the following parameters: $\gamma_0 = 1$, $\omega_1 = 0.3\,\text{rad/s}$, $\omega_2 = 30\,\text{rad/s}$, $T_{\text{owc}} = 20.9\,\text{s}$ (TB ≈ 99), and $t_w = 2\,\text{s}$. The strain, $\gamma(t)$, and strain rate, $\dot{\gamma}(t)$, are shown as a black dashed line and a black solid line, respectively. The Tukey window with $r = 10\%$ is illustrated as a gray dotted envelope. (b) and (c) Case 01: $\tau/T_{\text{owc}} = 0.06$; (b) output stress, $\sigma(t)$, is obtained by solving Eq. (12) numerically using $G_0 = 38\,\text{Pa}$ and $\tau = 1.19\,\text{s}$. The maximum expected stress (dotted lines) is $|\sigma_{\max}| = \gamma_0 G_0 = 38\,\text{Pa}$. (c) The amplitude spectrum $|G^*(\omega)|$ (blue filled circles) and phase $\delta(\omega)$ (blue hollow rectangles) are plotted over the frequency range of interest. CMM predictions for $|G^*(\omega)|$ and $\delta(\omega)$ are shown as black solid and dashed lines, respectively. The inset error plot highlights a negligible normalized mean absolute error (MAE) (<0.03%) between the numerical evaluations and CMM predictions. (d) and (e) Case 02: $\tau/T_{\text{owc}} = 1$; (d) $\sigma(t)$ (red solid line) is calculated by solving Eq. (12) with model parameters $G_0 = 38\,\text{Pa}$ and $\tau = T_{\text{owc}} = 20.9\,\text{s}$. The expected maximum stress value remains same, $|\sigma_{\max}| = 38\,\text{Pa}$ [indicated by dashes lines]. Gray-filled circles highlight the cycle-to-cycle deviation of the symmetry line of the stress signal. (e) The amplitude spectrum $|G^*(\omega)|$ (red filled circles) and phase $\delta(\omega)$ (red hollow rectangles) are compared with the CMM predictions for $G_0 = 38\,\text{Pa}$ and $\tau = 20.9\,\text{s}$, shown as black solid and dashed lines. The inset error plot shows an MAE of approximately 0.44% for $|G^*(\omega)|$ but 118.7% for $\delta(\omega)$.

As $G''(\omega)$ is a function of $\tan\delta(\omega)$ at very high frequencies, we see a greater impact of the noisy phase angle $\delta(\omega)$ on $G''(\omega)$. In general, spectral noise tends to disproportionately affect the subdominant component of the viscoelastic moduli —i.e., the component of the complex modulus with the smaller magnitude. For the Maxwell model, since $G''(\omega)$ decreases with increasing frequency, the reduced magnitude of the loss modulus at high frequencies makes it more

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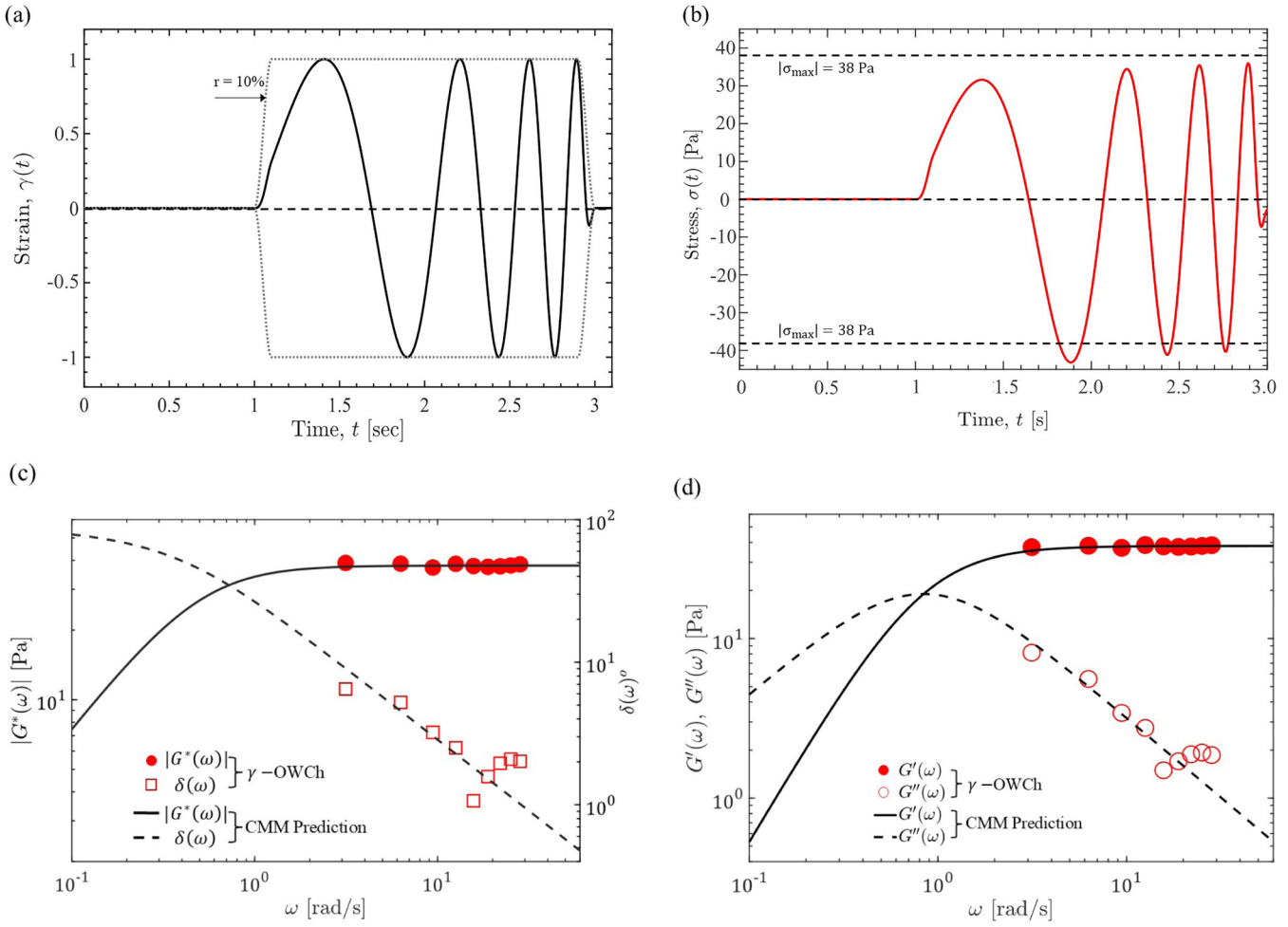


FIG. 4. (a) γ-OWCh waveform ($r = 10\%$) for a very short chirp with parameters $\omega_1 = 3$ rad/s, $\omega_2 = 30$ rad/s, $\gamma_0 = 1$, and $T_{owc} = 2$ s (TB ≈ 8.6). (b) Output stress, $\sigma(t)$, computed using Eq. (12); the expected maximum stress amplitude, $|\sigma_{max}| = G_0\gamma_0 = 38$ Pa, is indicated by dashed lines. (c) Comparison of the amplitude of the complex modulus, $|G^*(\omega)|$ (filled circles) and phase angle, $\delta(\omega)$ (hollow rectangles) obtained from the windowed chirp signal, with corresponding predictions from the CMM model shown as a solid black line and a dashed black line, respectively. (d) Comparison of the storage modulus $G'(\omega)$ and loss modulus $G''(\omega)$ predicted by the CMM (black solid and dashed lines) with the results obtained using the numerical investigation (filled and hollow circles, respectively).

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susceptible to the effects of noise. As a result, although noisy behavior propagates across the frequency spectrum, it is especially pronounced where the modulus values are the smallest. Moreover, additional computations show that by keeping τ and the frequency range ($\Delta\omega$) constant while increasing the chirp duration T_{owc}, both the time-bandwidth (TB) product improves and the ratio of τ/T_{owc} decreases (i.e., the locus of the point moves leftward along the abscissa toward the origin in the operating space of Fig. 1), resulting in a systematic reduction in the phase error and reduced noise in computation of $G''(\omega)$.

From these observations, we conclude that when the chirp duration T_{owc} is shortened until it becomes comparable to the material's inherent relaxation timescale τ, then the output stress signal $\sigma(t)$ loses perfect periodicity and symmetry in the time domain, disrupting the expected oscillatory response in which a trough of equal amplitude follows each peak.

Since our analysis is performed within the linear viscoelastic regime, the output stress response, $\sigma(t)$, at any present time t can be described using the hereditary integral formulation, commonly referred to as the Boltzmann superposition principle [28],

$$\sigma(t) = \int_{-\infty}^{t} G(t - t')\dot{\gamma}(t')dt', \tag{21}$$

where $G(t - t')$ is the material's relaxation modulus, characterizing how stress relaxes over time following a step-strain input. For the Maxwell model, the relaxation modulus takes the exponential form as

$$G(t - t') = G_0 e^{-(t-t')/\tau}. \tag{22}$$

This expression offers a direct interpretation of how the stress relaxes during a finite-duration experiment. For instance, if the experimental duration is set equal to the relaxation time, $1/e$, or approximately 36.8% of the initial stress resulting from a step-strain input will remain by the end of the experiment. By using a γ-OWCh waveform, we optimally window

the strain; however, a similar windowing effect observed in the output stress also depends on the time-dependent characteristics of the material's relaxation modulus $G(t)$. If the transient response, related to the material's inherent relaxation time (τ), decays within the chirp duration, the stress output is expected to closely follow the input windowing. For such materials, we can expect a periodic stress response where $\sigma(t = t - t_w) = \sigma(t = T_{owc} + t_w = 0)$, ensuring a clean viscoelastic spectrum in the Fourier domain [corresponding to Case 01 in Fig. 3: $\tau/T_{owc} = 0.06$, TB ≈ 99].

However, when τ is comparable to T_{owc}, the transient decay term propagates alongside the steady-state stress response. Thus, past deformations continue to influence the system due to the material's memory effect. As a result, even though the input strain waveform reaches zero at the end of the chirp signal due to windowing, the output stress does not immediately drop to zero. Instead, it takes some time to decay, depending on the value of τ. As a result, a finite residual stress value remains at the end of the chirp signal, creating a sharp edge in the periodic simulated data set. Replicating such a sharp edge in the Fourier domain requires a combination of sine and cosine waves across a wide range of frequencies. The contributions from these additional frequency components interfere with the original chirp frequency components, leading to a noisy spectrum [Case 02 in Fig. 3: $\tau/T_{owc} = 1$, TB ≈ 99 and Case 03 [Fig. 4]: $\tau/T_{owc} = 0.6$, TB ≈ 8.6]. These findings hold for chirp signals with any value of the TB product since the key dimensionless parameter in this context is τ/T_{owc}.

To tackle this situation, we explored tuning several different chirp experiment design parameters as well as post-processing parameters (e.g., waiting time before imposing chirp signal, t_w, sampling frequency, f_s, zero-padding [31], slew-rate $(d\omega(t)/dt)$, etc.). Finally, the modification we propose is to treat the initial phase, $\varphi_0 = \varphi(t - t_w = 0)$, of the input OWCh signal as a variable degree of freedom, rather than considering it a fixed value of zero ($\varphi_0 = 0$). This adjustment does not alter the "chirp rate" of the signal, meaning the frequency will still follow the same geometric relationship with time, as described by $\omega(t - t_w)$ [Eq. (2)]. Importantly, this additional constant phase term can be incorporated directly within the ARES-G2 rheometer's Arbitrary Wave functionality. The character limit of the waveform-definition interface is sufficient to specify this additional tuning parameter.

By introducing a finite value of φ_0 in the time-domain signal, we can precisely control its temporal shift. Consequently, this phase adjustment modifies the strain by altering the position in time of successive peaks and troughs, effectively shifting the timing of different frequency components within the chirp signal. The key idea behind this phase tuning is to align $\dot{\gamma}(t')$ relative to the relaxation modulus kernel $G(t - t')$ such that the convolution integral balances more symmetrically. This alignment is achieved by optimizing the initial phase φ_{opt} to minimize the residual stress at the end of the chirp signal $(t = T_{owc} + t_w)$.

Figures 5(a) and 5(b) compare two $\gamma - OWCh$ waveforms that differ only in their initial phase: $\varphi_0 = 0°$ and $\varphi_0 = 90°$, along with their corresponding stress responses. The stress and strain signals are normalized for comparison $(\hat{\gamma}(t) = \gamma(t)/\gamma_0,\ \hat{\sigma}(t) = \sigma(t)/\sigma_{max})$. Setting a finite nonzero initial phase ($\varphi_0 = 90°$) simply repositions the oscillations within the same window and chirp sweep. In this case, the stress ends closer to zero and is more periodic and symmetric than in the $\varphi_0 = 0°$ case. This reduces end effects and spectral leakage, yielding a cleaner spectrum. It is important to note that this phase shift does not alter the power spectrum of the input; it adds a constant phase offset that improves the periodicity and symmetry alignment of the stress signal in time. Consequently, the final quantities of interest in OWCh rheometry—namely, the magnitude of the complex modulus $|\tilde{\sigma}(\omega)/\tilde{\gamma}(\omega)|$ and the phase difference $\delta(\omega)^{\circ}_{\sigma} - \delta(\omega)^{\circ}_{\dot{\gamma}}$ — are unaffected by a uniform phase shift. This is because both the

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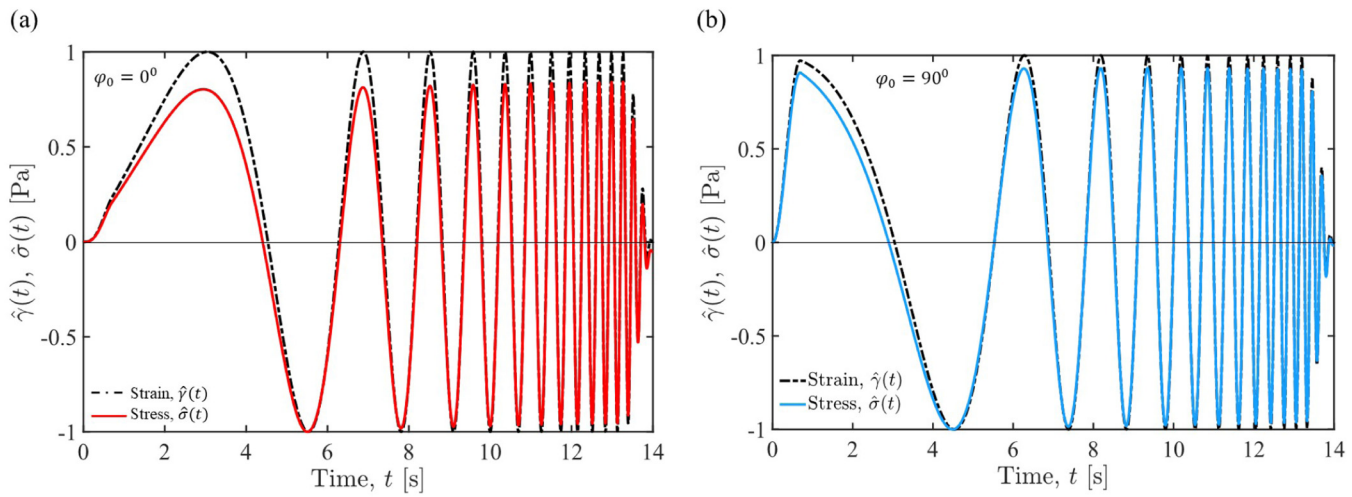


FIG. 5. Example of γ-OWCh waveforms with different initial phase angles: $\varphi_0 = 0°$ and $\varphi_0 = 90°$. Both γ-OWCh waveforms have the following parameters: $\omega_1 = 0.3$ rad/s, $\omega_2 = 30$ rad/s, $T_{owc} = 14$ s, TB ≈ 66, and $r = 10\%$. (a) With $\varphi_0 = 0°$, the stress response does not return to zero at the end of the chirp and appears asymmetric. (b) Introducing a finite initial phase ($\varphi_0 = 90°$) realigns the crest, trough, and zero crossings of the input strain signal. This adjustment causes the output stress to be close to zero and symmetric about the centerline axis.

stress and strain signals experience the same temporal translation, and hence their magnitude and relative phase relationships remain unchanged.

To further investigate the impact of the optimal initial phase shift—particularly when τ/T_{owc} is comparable to or greater than unity—we generated contour plots showing the variation of error with respect to φ_0 for different τ/T_{owc} ratios. Because the classical Maxwell model is used for this analysis, the error was calculated using the loss modulus $G''(\omega)$, which is the subdominant component and more significantly impacted by the noisy phase spectrum.

For each value of φ_0, the MAE was calculated using the following expression:

$$\text{MAE, } \varepsilon = \frac{1}{N}\sum_{i=1}^{N}\left|\frac{G''_{\text{Simulated}}(\omega) - G''_{\text{CMM}}(\omega)}{G''_{\text{CMM}}(\omega)}\right| \times 100\,\%, \tag{23}$$

where

$$G''_{CMM}(\omega) = G_0\frac{(\tau\omega)}{1+(\tau\omega)^2}. \tag{24}$$

For all simulations, the elastic modulus ($G_0 = 38$ Pa) was kept constant. The frequency range was set from $\omega_1 = 0.3$ rad/s to $\omega_2 = 30$ rad/s, and the chirp duration was defined as $T_{owc} = 2\pi/\omega_1$ (TB $\approx$ 99). The material relaxation time, τ, was varied to achieve τ/T_{owc} ratios ranging from 0.1 to 2.

The resulting contour plot is shown in Fig. 6(a). To highlight regions of high accuracy, we marked areas where the error is less than 5% with light gray dashed lines. From this plot, several important conclusions can be drawn:

- When $\tau/T_{owc} \ll 1$, the impact of the initial phase shift on the error, ε, is negligible. In this regime, all values of φ_0 result in high accuracy data of $G'(\omega)$, $G''(\omega)$ (error $\leq 5\%$), indicating that the choice of φ_0 does not influence the outcome significantly.
- With increasing values of the τ/T_{owc} ratio, there are two narrow regions of optimal φ_0 separated by 180° or π rad [gray dashed lines in Fig. 6(a)] with the smallest possible error.
- The regions of an optimal phase angle for minimizing error converge to a nearly constant value as τ/T_{owc} becomes large.

This plot is particularly significant in experimental scenarios where chirp signals are used to track the evolving viscoelastic spectra of materials as a function of temperature (K). As the temperature changes, the material's inherent relaxation time $\tau(K)$ varies, thus altering the ratio $\tau(K)/T_{owc}$, assuming that the chirp duration, T_{owc}, remains constant. Based on this consideration and the general shape of Fig. 6(a), the optimal phase angle of the input signals that will minimize error may also change. These insights are equally applicable to mutating materials, such as those undergoing a phase transition from a liquid-like to a solid-like state or gelling materials that lead to increasingly viscoelastic behavior.

We also illustrate the variation in MAE for a specific case of $\tau/T_{owc} = 1$ in the 2D plot shown in Fig. 6(b). For this particular case, there are two distinct optimal initial phase values, $\varphi_0 = \varphi_{opt} = -80.5°$ and $+99.5°$ (180° (or π rad) apart as per the periodic nature of the Fourier transform), that result in exactly 0% error for the given chirp-design parameters. The waveforms of input strain, $\gamma(t)$ for $\varphi_0 = 0°$ and $\varphi_0 = \varphi_{opt} = -80.5°$, as well as the output stress, $\sigma(t)$, and the corresponding viscoelastic properties are presented in Fig. 7. Figure 7(a) shows that adjusting the input γ-OWCh waveform to the optimal phase angle, $\varphi_{opt} = -80.5°$, forces the signal to rapidly grow to a higher initial amplitude and modifies the positioning of the oscillations (peaks, troughs, and zero crossings). This shift helps regulate the output stress by reducing the impact of the transient stress response, controlling the placement of peaks, troughs, zero crossings, and

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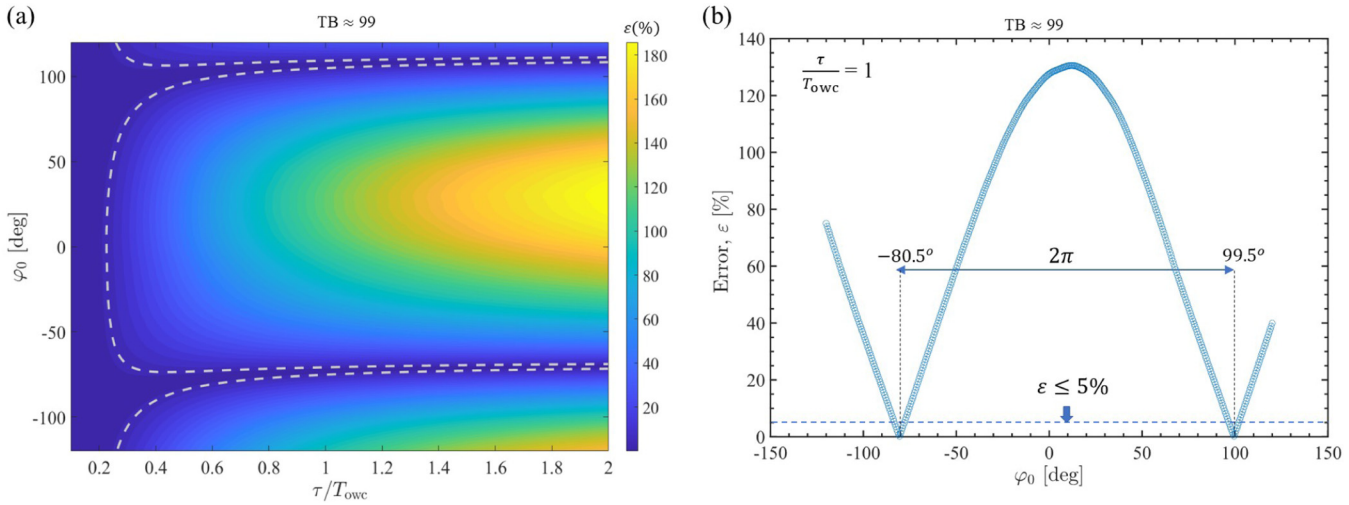


FIG. 6. (a) Contour plot of the error, $\varepsilon(\varphi_0, \tau/T_{owc})$. The chirp parameters used are $\omega_1 = 0.3$ rad/s, $\omega_2 = 30$ rad/s, $T_{owc} = 20.9$ s, TB $\approx$ 99, and $r = 10\%$. For the CMM parameters, $G_0 = 38$ Pa is kept fixed and τ varies from 2.09 to 41.8 s, resulting in a τ/T_{owc} ratio ranging from 0.1 to 2. The dashed lines indicate regions where the error, $\varepsilon \leq 5\%$. (b) Error plot corresponding to $\tau/T_{owc} = 1$. The dashed line highlights the range of phase angles where $\varepsilon \leq 5\%$. For the specified chirp parameters, two optimal phase angles (−80.5°, 99.5°) separated by 180° result in a zero percent mean absolute error ($\varepsilon = 0\%$).

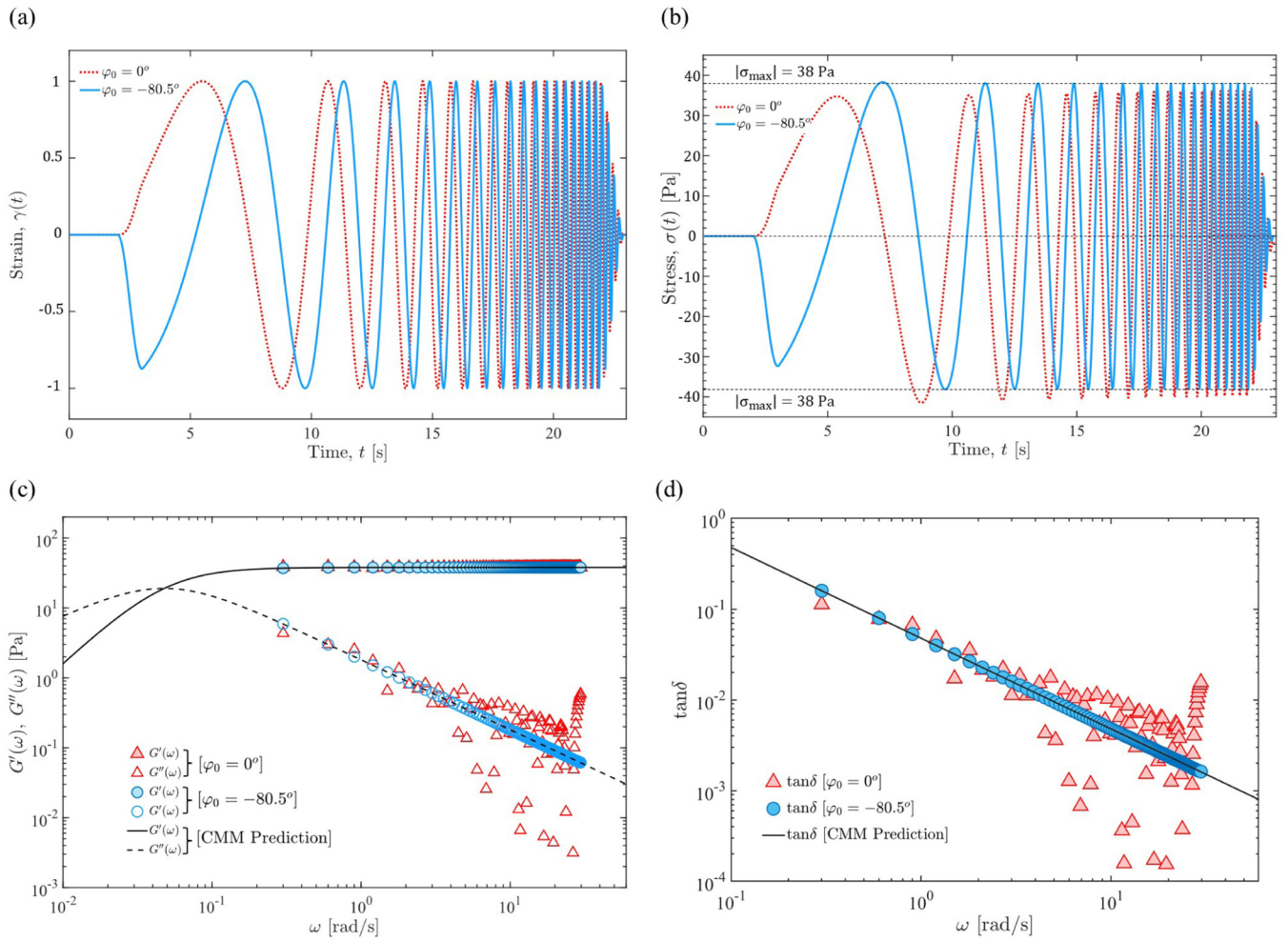


FIG. 7. Data obtained by numerically solving Eq. (12) for an input γ-OWCh waveform. (a) Both input γ-OWCh signals have the following parameters: $\omega_1 = 0.3$ rad/s, $\omega_2 = 30$ rad/s, $\gamma_0 = 1$, $T_{owc} = \tau = 20.9$ s (TB ≈ 99), r = 10% and $t_w = 2$ s but different initial phases: $\varphi_0 = 0^\circ$ (red dotted line), and optimal phase, $\varphi_0 = -80.5^\circ$ (blue solid line). (b) Output stress waveforms, $\sigma(t)$ corresponding to $\varphi_0 = 0^\circ$ and $\varphi_0 = -80.5^\circ$, are shown using the same color coding as in (a). The optimal initial phase shift ($\varphi_0 = -80.5^\circ$) results in symmetric and periodic output stress. (c) Comparison of viscoelastic moduli, $G'(\omega)$ and $G''(\omega)$, obtained from windowed chirp signals with different initial phase shifts. Data for $\varphi_0 = 0^\circ$ are represented as filled triangles ($G'(\omega)$) and open triangles ($G''(\omega)$), while data for $\varphi_0 = -80.5^\circ$ are shown as filled circles ($G'(\omega)$) and open circles, ($G''(\omega)$). Reference predictions from the CMM for $G'(\omega)$ and $G''(\omega)$ are shown as black solid and black dotted lines, respectively. (d) Comparison of the phase angle, $\tan\delta(\omega) = G''(\omega)/G'(\omega)$ for $\varphi_0 = 0^\circ$ and $\varphi_0 = -80.5^\circ$. The same color coding and symbols as in (c) are used.

ensuring a more symmetric and periodic behavior in the output stress signal [Fig. 7(b)]. A direct comparison of the viscoelastic moduli, $G'(\omega)$, $G''(\omega)$, obtained for different phase shifts with the ground truth (i.e., the classical Maxwell model predictions), is presented in Fig. 7(c). Similarly, Fig. 7(d) compares the phase angle, $\delta(\omega)$, obtained for different phase shifts with the ground truth. The results demonstrate excellent agreement of the viscoelastic moduli (as well as the phase angle) that are computed with the optimal phase shift with the ground truth. A similar analysis was conducted for the other limiting case considered earlier, i.e., TB ≈ 8.6 [Case: 03], with results presented in the Appendix. The conclusions remain consistent for this case as well.

Up to this point, all the analyses discussed have been based on a simple two-parameter classical Maxwell model with a single dominant relaxation timescale. However, real-world complex fluids typically exhibit broad relaxation spectra. To account for this, we extend our analysis using the Fractional Maxwell model, which has proven to be highly effective in compactly and accurately capturing the linear rheological behavior of various viscoelastic materials [32–34]. The detailed analysis is presented in Subsection II B.

B. Fractional Maxwell model

The FMM consists of a series arrangement of two fractional elements, commonly referred to as spring-pots [35,36], enabling it to capture the relaxation dynamics of materials with broad relaxation spectra compactly and accurately using only four parameters. The one-dimensional constitutive equation of the FMM in a differential form is given by [32,33,37]

$$\sigma(t) + \frac{\mathbb{V}}{\mathbb{G}} \frac{d^{\alpha-\beta}}{dt^{\alpha-\beta}} \sigma(t) = \mathbb{V} \frac{d^{\alpha}\gamma(t)}{dt^{\alpha}}. \tag{25}$$

In Eq. (25), $\mathbb{V}$, $\mathbb{G}$ are quasi-properties that set the magnitude of the material response, and α, β are the power-law exponents that characterize the dynamical response of the spring-pots. We assume $0 \leq \beta < \alpha \leq 1$ without loss of generality [32]. The quasi-properties $\mathbb{V}$ and $\mathbb{G}$ have units Pa $\cdot$ s$^\alpha$ and

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$\mathrm{Pa}\cdot\mathrm{s}^{\beta}$, respectively. In the limiting case of $\alpha = 1$ and $\beta = 0$, the FMM simplifies to the simple Maxwell model. Considering the physical dimensions of $\mathbb{V}$ and $\mathbb{G}$, we can identify a single characteristic relaxation time (with SI units of seconds) of the FMM to be

$$\tau_c = (\mathbb{V}/\mathbb{G})^{\frac{1}{\alpha-\beta}}. \tag{26}$$

This represents the appropriate average timescale of the broad underlying continuous relaxation spectrum [34,37].

Expressions for the storage modulus, $G'(\omega)$, and the loss modulus, $G''(\omega)$, of the FMM can be written as follows [33,37]:

$$\frac{G'(\omega)}{G_c} = \frac{(\omega\tau_c)^{\alpha}\cos(\pi\alpha/2) + (\omega\tau_c)^{2\alpha-\beta}\cos(\pi\beta/2)}{1 + 2(\omega\tau_c)^{\alpha-\beta}\cos(\pi(\alpha-\beta)/2) + (\omega\tau_c)^{2(\alpha-\beta)}}, \tag{27}$$

$$\frac{G''(\omega)}{G_c} = \frac{(\omega\tau_c)^{\alpha}\sin(\pi\alpha/2) + (\omega\tau_c)^{2\alpha-\beta}\sin(\pi\beta/2)}{1 + 2(\omega\tau_c)^{\alpha-\beta}\cos(\pi(\alpha-\beta)/2) + (\omega\tau_c)^{2(\alpha-\beta)}}, \tag{28}$$

where G_c represents the characteristic modulus of the FMM, which can be written in terms of the quasi-property $\mathbb{V}$ and the characteristic time constant τ_c as $G_c = \mathbb{V}\tau_c^{-\alpha}$ with the SI unit (Pa).

To investigate the impact of the initial phase shift (φ_0) of the chirp signal on accurately determining the viscoelastic spectrum of materials with a broad relaxation time distribution, we use the FMM parameters determined from regression to the experimental storage and loss modulus data of an 8.5 wt.% poly(isobutylene) in hexadecane [also referred to compactly as a PIB solution], as presented by Geri *et al.* [7] in Fig. 5 of their study. The parameters are: $\alpha = 1, \mathbb{V} \equiv \eta_0 = 18\,\mathrm{Pa}\cdot\mathrm{s}$, $\beta = 0.36$, $\mathbb{G} = 50\,\mathrm{Pa}\cdot\mathrm{s}^{\beta}$; thus, $\tau_c = 0.20\,\mathrm{s}$. The value of the power-law exponent, $\alpha = 1$, represents a special limiting case of the FMM model—one spring-pot reduces to a purely viscous dashpot with $\mathbb{V} \equiv \eta_0\,[\mathrm{Pa}\cdot\mathrm{s}]$, and this gives rise to a finite zero-shear rate viscosity. This three-parameter model is known as the Fractional Maxwell Liquid (FML) model [34,38].

The output stress $\sigma(t)$ is simulated using Eq. (21). For the fractional Maxwell model, the relaxation modulus, $G(t-t')$, can be expressed as follows:

$$G(t-t') = \mathbb{G}t^{-\beta}E_{\alpha-\beta,1-\beta}\left(-(t/\tau_c)^{\alpha-\beta}\right), \tag{29}$$

where $E_{a,b}$ is the two-parameter Mittag-Leffler function,

$$E_{a,b} = \sum_{k=0}^{\infty}\frac{z^k}{\Gamma(ak+b)} \quad (a>0,\ b>0). \tag{30}$$

This functional form accurately captures the short and long timescale power-law behavior of a range of viscoelastic materials [39].

We begin by verifying whether our first conclusion above, that for $\tau_c/T_{\mathrm{owc}} \ll 1$, any finite value of φ_0 provides the same level of accuracy as a zero initial phase shift ($\varphi_0 = 0°$), remains valid for the FMM. To test this, we conducted numerical simulations using an input γ-OWCh signal with a frequency range from $\omega_1 = 0.3\,\mathrm{rad/s}$ to $\omega_2 = 30\,\mathrm{rad/s}$ and a chirp duration of $T_{\mathrm{owc}} = (2/3)\times(2\pi/\omega_1) = 14\,\mathrm{s}$, resulting in a time-bandwidth product of TB ≈ 66 (all parameters are chosen similar to Geri *et al.* [7]). The ratio of the characteristic relaxation of the fluid to the duration of the chirp signal is $\tau_c/T_{\mathrm{owc}} = 0.01 \ll 1$. The value of the tapering parameter is taken to be $r = 10\%$. We considered two initial phase shifts, $\varphi_0 = 0°$ and $\varphi_0 = 90°$.

The results of the numerical simulation are compared with the ground truth, i.e., the analytical expressions for $G'(\omega)$ and $G''(\omega)$ [Eqs. (27) and (28)]. The resulting plots of $G'(\omega)$ and $G''(\omega)$ are shown in Fig. 8(a), demonstrating that both chirp signals, regardless of the initial phase shift [$\varphi_0 = 0°$ or $\varphi_0 = 90°$], achieve the same level of accuracy when compared to the ground truth. However, using chirp signals with a higher value of initial phase shift ($\varphi_0 = 90°$) offers an additional advantage—it enhances the SNR, particularly toward the start and end of the chirp signal, where the amplitude gradually increases from zero and decreases toward zero due to the tapering effect of the Tukey window. This may be anticipated to be especially beneficial in mitigating the impact of the inevitable noise floor present in any experimental procedure.

Next, we test the second conclusion derived from our analysis with the CMM, which asserts that when $\tau/T_{\mathrm{owc}} \not\ll 1$, there are two optimal phase angles φ_{opt} (180° apart) that minimize the error in capturing accurate viscoelastic moduli. To validate this, we considered a short chirp with TB ≈ 8.6 designed using $\omega_1 = 3\,\mathrm{rad/s}$, $\omega_2 = 30\,\mathrm{rad/s}$, $T_{\mathrm{owc}} = 2\,\mathrm{s}$, $r = 10\%$. In this case, the ratio $\tau/T_{\mathrm{owc}} = 0.1$. The resulting viscoelastic spectra are presented in Fig. 8(b). As shown in the figure, by using the optimal phase angle $\varphi_{\mathrm{opt}} = 50°$, we can effectively utilize chirp signals with a time-bandwidth product as low as TB < 10 and $\tau/T_{\mathrm{owc}} \not\ll 1$ while still achieving good accuracy.

III. EXPERIMENTAL INVESTIGATION

To experimentally investigate the influence of φ_0 on the OWCh protocol, we consider two categories of viscoelastic materials:

The first category includes stable (or very slowly mutating) materials that satisfy the time-translation invariant (TTI) principle within the experimental window ($Mu \ll 1$). For this category, we analyze two reference fluids: (i) A micellar solution composed of cetylpyridinium chloride (CPyCl), sodium salicylate (NaSal), and sodium chloride (NaCl) in deionized water, with concentrations of 100:60:33 mM (CPyCl:NaSal:NaCl) and (ii) a high-strength acrylic adhesive 300 LSE obtained from 3M.

For the second category, we consider materials that are not time-translation invariant within the experimental timeframe. For this case, we consider a UV-curing material, which exhibits rapid evolution in its viscoelastic relaxation spectrum during the curing process.

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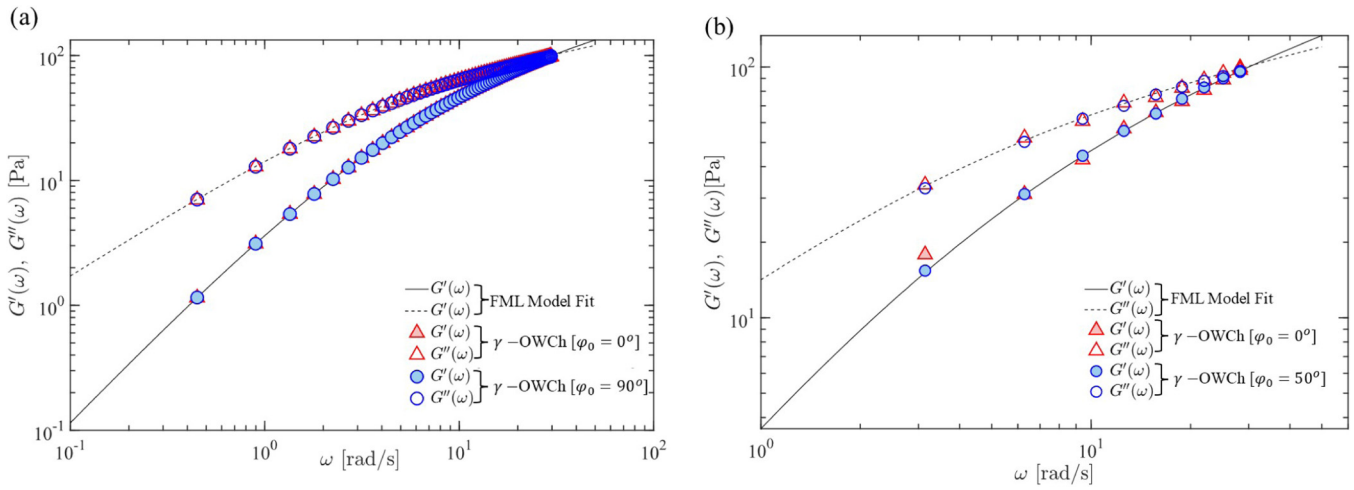


FIG. 8. Data presented are obtained by numerically solving Eq. (25) for an input γ-OWCh waveform. The parameters of the FML model used for investigation are: $\alpha = 1$, $\mathbb{V} \equiv \eta_0 = 18\,\text{Pa}\cdot\text{s}$, $\beta = 0.36$, $\mathbb{G} = 50\,\text{Pa}\cdot\text{s}^\beta$, $\tau_c = 0.20\,\text{s}$ [Fig. 5 of the original paper by Geri *et al.* [7]; Geri et al., Phys. Rev. X **8**, 041042 (2018). Copyright 2018 Author(s), licensed under a Creative Commons Attribution 4.0 International License]. The analytical prediction of the FML model (Eqs. (27) and (28) with $\alpha = 1$) are shown using black solid ($G'(\omega)$) and black dashed ($G''(\omega)$) lines over the frequency range of interest. (a) $\tau_c/T_{\text{owc}} \ll 1$, TB ≈ 66: Data corresponding to $\varphi_0 = 0^\circ$ are represented as filled triangles ($G'(\omega)$) and open triangles ($G''(\omega)$), while data for $\varphi_0 = 90^\circ$ are shown as filled circles ($G'(\omega)$) and open circles ($G''(\omega)$). The results demonstrate that both phase shifts yield comparable accuracy relative to the analytical solution. (b) $\tau_c/T_{owc} \not\ll 1$, TB ≈ 8.6: $G'(\omega)$ and $G''(\omega)$ data using the optimal phase shift $\varphi_{\text{opt}} = 50^\circ$ (blue filled and blue hollow circles) show improved accuracy compared to the results obtained with $\varphi_0 = 0^\circ$ (red filled and red hollow triangles) when compared to the analytical predictions.

A. Wormlike micellar solution

Rheometric measurements were performed by using a TA Instruments ARES-G2 Rheometer, with a bottom Peltier plate, and an upper 50 mm 1.117° stainless steel cone plate with truncation gap 55.5 μm.

Figure 9(a) illustrates the dynamic moduli, $G'(\omega)$, $G''(\omega)$, of the micellar sample at a temperature of K = 25°C. Each input strain-controlled chirp (γ-OWCh) was designed considering the frequency range, $\omega_1 = 3\,\text{rad/s}$, $\omega_2 = 30\,\text{rad/s}$, with a duration of 2 s, resulting in a TB product of 8.6. Two initial phase angles were considered: $\varphi_0 = 0^\circ$ and $\varphi_0 = -90^\circ$. A waiting time of $t_w = 1$ s is imposed at the beginning of the chirp signal to ensure that the strain signal starts at a zero value. The input strain amplitude was set at $\gamma_0 = 15\%$, which is within the linear viscoelastic regime ($\gamma_0 \leq 25\%$), as confirmed independently by an oscillatory strain-amplitude sweep experiment at a frequency of 1 rad/s. As shown in Fig. 9(a), utilizing a γ-OWCh signal with a finite initial phase shift provides improved measurements of $G'(\omega)$ and $G''(\omega)$ compared to a γ-OWCh signal with zero initial phase shift. To further evaluate the accuracy of viscoelastic data obtained using a short time-bandwidth product chirp (TB ≈ 8.6), we compared it with results from another γ-OWCh

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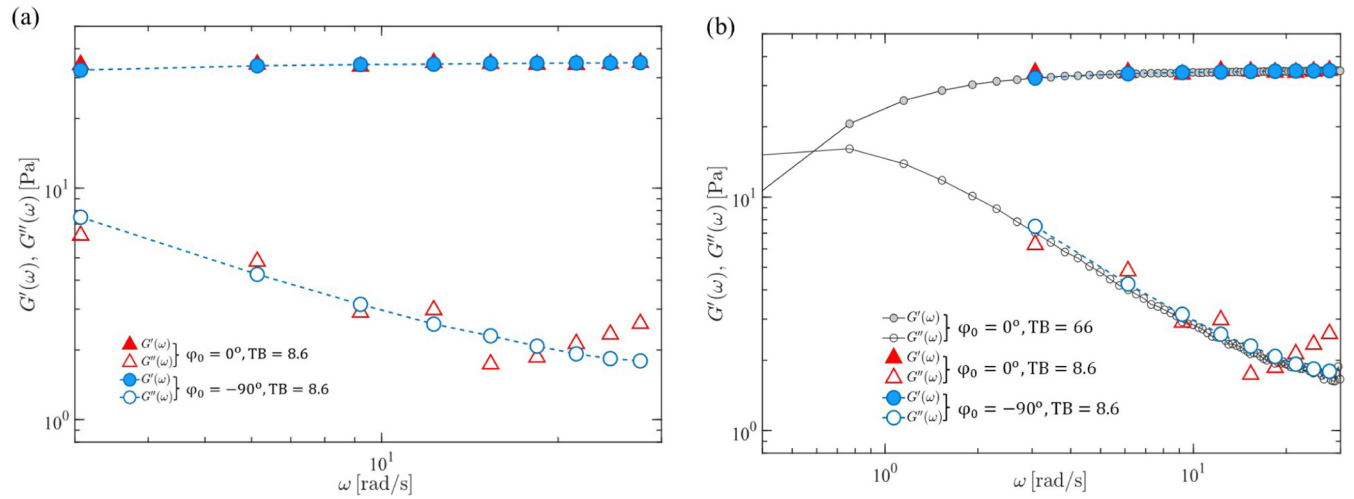


FIG. 9. (a) Viscoelastic moduli of the CPyCl micellar solution. Comparison between results measured using an optimally windowed chirp signal with zero initial phase shift, $\varphi_0 = 0^\circ$ ($G'(\omega)$, red filled triangles; $G''(\omega)$, red open triangles), and with $\varphi_0 = -90^\circ$ ($G'(\omega)$, blue filled circles; $G''(\omega)$, blue open circles). (b) Comparison of the viscoelastic spectra obtained using short chirps with TB ≈ 8.6 for $\varphi_0 = 0^\circ$ and $\varphi_0 = -90^\circ$ against the reference spectrum obtained using a higher time-bandwidth product chirp TB ≈ 66 ($\tau_c/T_{\text{owc}} \ll 1$) with $\varphi_0 = 0^\circ$ represented by filled gray circles ($G'(\omega)$) and open gray circles ($G''(\omega)$). The symbols and color coding for short TB product chirps are identical to those in (a). The data obtained using γ-OWCh with TB ≈ 8.6 and $\varphi_0 = -90^\circ$ show excellent agreement with the $G'(\omega)$ and $G''(\omega)$ measurements using γ-OWCh with TB ≈ 66 and $\varphi_0 = 0^\circ$.

signal designed with a higher time-bandwidth product (TB ≈ 66). The chirp-design parameters for this signal are $\omega_1 = 0.3\,\text{rad/s}$, $\omega_2 = 30\,\text{rad/s}$, $T_{\text{owc}} = 14\,\text{s}$, $r = 10\%$.

As shown in Fig. 9(b), the micellar solution exhibits a characteristic crossover frequency, $\omega_c = 0.6\,\text{rad/s}$, and a corresponding relaxation time, $\tau_c = 1.7\,\text{s}$. Therefore, the chirp duration of 14 s satisfies the condition $\tau/T_{\text{owc}} \ll 1$, ensuring reliable spectral measurements for any values of the initial phase shift (φ_0). Figure 9(b) also demonstrates that the viscoelastic moduli $G'(\omega)$ and $G''(\omega)$ obtained using the low-TB chirp (TB ≈ 8.6 and $\varphi_0 = -90°$) closely match those from the high-TB chirp (TB ≈ 66 and $\varphi_0 = 0°$), confirming the accuracy of the low-TB approach when combined with an optimal initial phase condition ($\varphi_0 = -90°$).

B. 3M™ high-strength acrylic adhesive 300 LSE

Rheometric measurements were performed at room temperature (25° C) by using an ARES-G2 Rheometer (TA Instruments, New Castle, DE), with a bottom Peltier plate, and an upper 8 mm stainless steel parallel plate with a gap of 0.85 mm.

In Fig. 10(a), we present the storage and loss moduli of the adhesive sample, measured using the same chirp design parameters as in the micellar solution study: TB ≈ 8.6. We consider two initial phase shifts: $\varphi_0 = 0°$ and $\varphi_0 = 45°$. The input strain amplitude was set at $\gamma_0 = 10\%$, which is within the linear viscoelastic regime ($\gamma_0 \leq 30\%$), as confirmed by an independent oscillatory strain-amplitude sweep experiment at a frequency of 6 rad/s. As shown in Fig. 10(a), utilizing a γ-OWCh signal with a finite initial phase shift provides smoother measurements of $G'(\omega)$ and $G''(\omega)$ compared to a γ-OWCh signal with zero initial phase shift.

To assess the accuracy of the chirp-based measurements, in Fig. 10(b), we compare these same data (obtained using γ-OWCh with $\varphi_0 = 0°$ and 45°) with reference results from a conventional DFS experiment (black solid and dashed lines). The conventional DFS test required nearly 10 min to complete, including two delay cycles at each frequency before measurement. In the overlapping frequency window, i.e., $\omega_1 = 3\,\text{rad/s}$ and $\omega_2 = 30\,\text{rad/s}$, the conventional DFS measurement required 221 s for obtaining six data points, whereas the OWCh method captured nine data points in just 2 s. The measurements obtained with a finite initial phase shift ($\varphi_0 = 45°$) show excellent agreement with the $G'(\omega)$ and $G''(\omega)$ values obtained from the DFS test, highlighting that phase optimization significantly improves the reliability of chirp-based rheometry for accurate viscoelastic characterization.

C. UV-curing material

Experiments on the UV-curable acrylate system were performed at 20 °C using a 20 mm disposable acrylate top plate. Measurements were performed on a TA Instruments ARES-G2 rheometer with a sample gap of 500 μm. UV illumination was provided by a 10 W/cm^2 Excelitas Omnicure LX500 LED source operated at 60% power. Before each test, the lamp output was calibrated for transmittance variations through the top plate. To monitor the cross-linking process—characterized by the transition from a mobile liquid to a highly cross-linked state—100 consecutive OWCh signals were applied over a total duration of 1080 s. Each input OWCh signal was designed considering the frequency range, $\omega_1 = 1.5\,\text{rad/s}$, $\omega_2 = 120\,\text{rad/s}$, with a duration of 2.8 s, resulting in a time-bandwidth product TB = 52. This duration was chosen so that at each time step, the condition $Mu \ll 1$ was met. Two initial phase angles were considered: $\varphi_0 = 0°$ and $\varphi_0 = -85°$. A waiting time of $t_w = 2.2\,\text{s}$ is imposed at the beginning of the chirp signal to ensure that the input strain signal settles from the previous chirp and starts at zero value. The input strain amplitude, γ_0, was chosen within the LVE limit. Since the extent of the LVE regime evolves during the *in situ* cross-linking, γ_0 was

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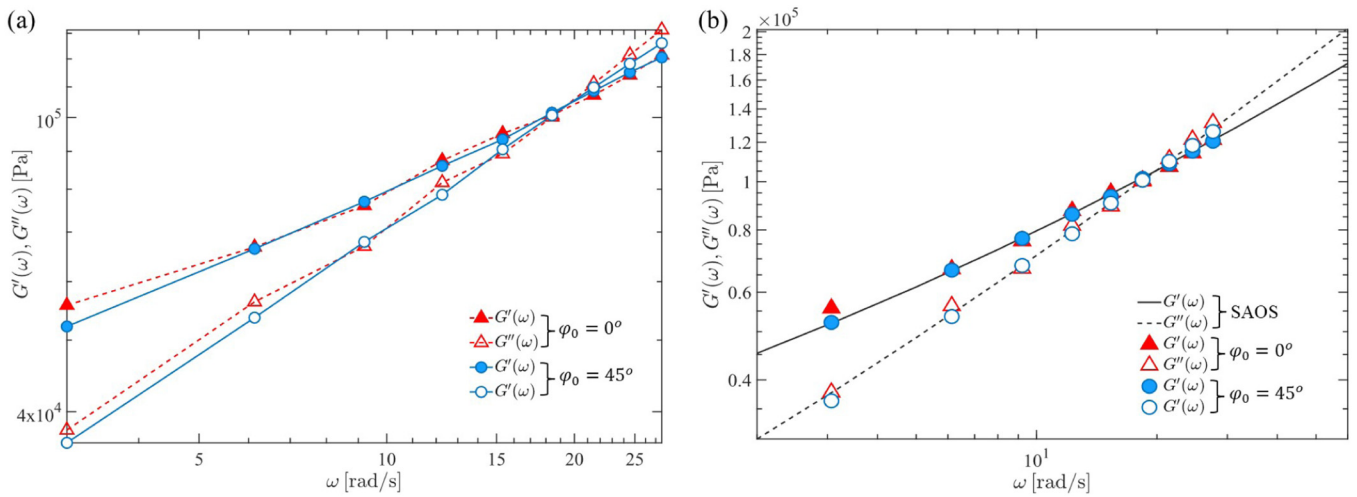
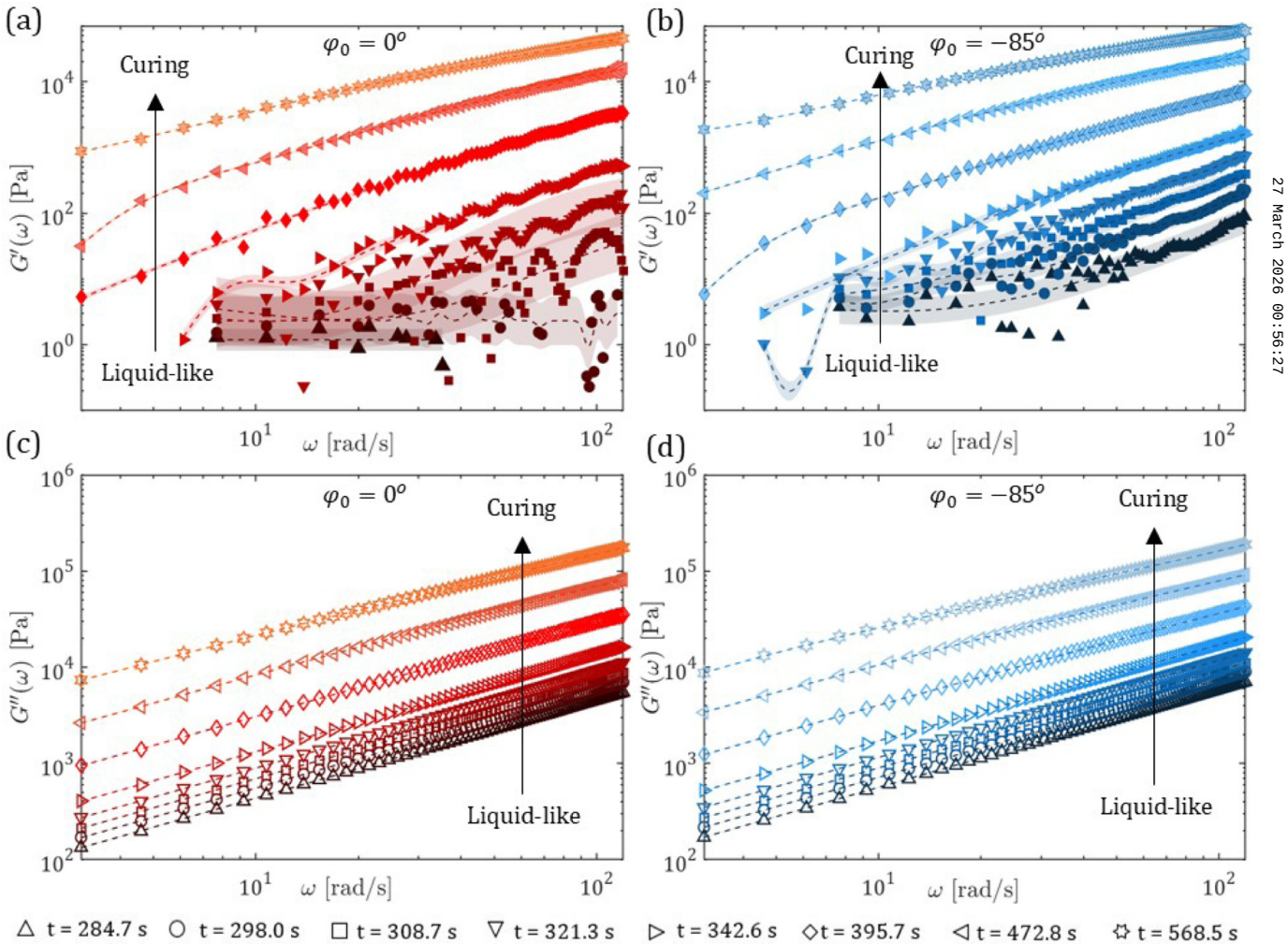


FIG. 10. Viscoelastic moduli of 3M™ high-strength acrylic adhesive 300 LSE. (a) Comparison between results measured using γ-OWCh signals with initial phase shift, $\varphi_0 = 0°$ ($G'(\omega)$, red filled triangles; $G''(\omega)$, red open triangles), and $\varphi_0 = 45°$ ($G'(\omega)$, blue filled circles; $G''(\omega)$, blue open circles). (b) Comparison of γ-OWCh results obtained at different phase angles [$\varphi_0 = 0°$ and $\varphi_0 = 45°$; the symbols and color code used are identical to those in (a)] with ground truth observations (obtained from discrete frequency sweep (SAOS) measurements taking almost 10 min to complete, including a waiting time equal to two delay cycles).

adjusted accordingly. A Tukey window with $r = 6\%$ was used for the chirp signal design. Notably, this material was also recently examined in detail by Perego and co-authors [14]. In their study, they described how the frequency-averaged mutation time, $\overline{Mu}(t)$ [refer to Eq. (15) and Fig. 7 of the original paper], evolves throughout the cross-linking process. They also outlined the design of OWCh experiments that capture the changing viscoelastic spectrum during cross-linking while ensuring the condition $\overline{Mu}(t) \ll 1$ is satisfied. For more information, readers are referred to their study. Here, we focus solely on the role of the initial phase in the accurate determination of the viscoelastic spectra.

Figure 11 presents a comparative analysis of the evolution of the storage modulus $G'(\omega)$ and the loss modulus $G''(\omega)$ at various stages throughout the UV-curing process. Data are shown starting from time $t = 284.7$ s, as the material remains in an uncured, liquid state prior to this point. In the early phase ($t < 284.7$ s), the output stress signal is extremely weak and limited by the rheometer's inherent noise floor, which hinders accurate determination of the viscoelastic spectrum. We present results from eight distinct OWCh experiments, capturing the material's transition from a dissipation-dominated (liquid-like) initial regime into a progressively more elastic solid material as the cross-link density increases (up to a maximum time, $t = 568.5$ s). Figures 11(c) and 11(d) focus on the evolution of the loss modulus, $G''(\omega, t)$, with time. Because $G''(\omega, t)$ maintains relatively higher values across all stages, its spectrum remains clean regardless of the initial phase angle of the chirp signal. One of the noticeable trends in Figs. 11(a) and 11(b) is that during the early stages of cross-linking, the storage modulus, $G'(\omega, t)$—being the subdominant component—shows greater susceptibility to noise introduced in the Fourier domain arising from the lack of chirp signal periodicity and symmetry. This is especially evident in the initial liquid-like state, where the $G'(\omega)$ values are significantly lower than $G''(\omega)$.



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FIG. 11. Time-resolved rheological characterization of the UV-curable acrylate system undergoing *in situ* cross-linking using two different initial phase angles, $\varphi_0 = 0°$, $\varphi_0 = -85°$. Gaussian Process Regression (GPR)-derived posterior means are shown as dashed lines, with shaded envelopes representing the corresponding uncertainty bounds. (a) and (b) Comparison of the evolution of the storage modulus, $G'(\omega, t)$, at various stages throughout the cross-linking process. Chirp signals with an initial phase shift of $\varphi_0 = -85°$ produce less noisy $G'(\omega)$ results, particularly during the early, liquid-like stage than those with $\varphi_0 = 0°$. As the material becomes increasingly cross-linked over time, both phase angles yield spectra of comparable quality. (c) and (d) Evolution of the loss modulus, $G''(\omega, t)$. Note the difference in ordinate scales. It is important to note that the loss modulus $G''(\omega)$ remains significant throughout all stages of the cross-linking process. As a result, its spectrum consistently appears clear and is less influenced by the initial phase of the input chirp signal.

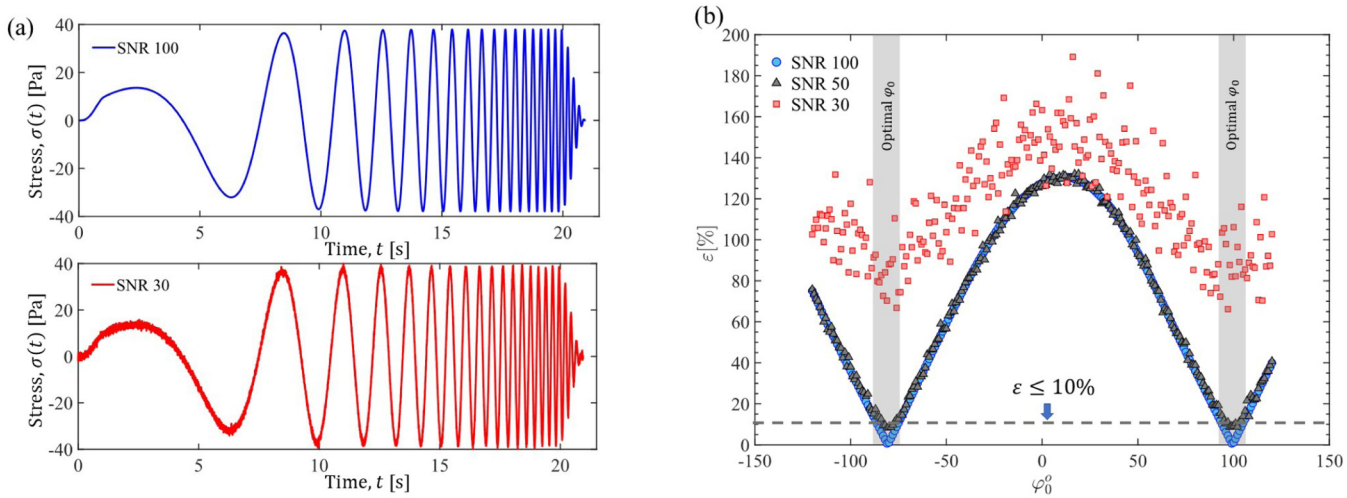


FIG. 12. Effect of high-frequency noise on determination of the optimal phase angle φ_0 when $\tau/T_{owc} = 1$. Data presented are obtained numerically by solving the classical Maxwell model constitutive equation. (a) Gaussian noise is added to the output stress resulting from an input chirp signal with parameters $\omega_1 = 0.3$ rad/s, $\omega_2 = 30$ rad/s, $\gamma_0 = 1$, $T_{owc} = \tau = 20.9$ s (TB $\approx$ 99), r = 10%. (b) The ordinate represents the normalized mean absolute error (ε) calculated using Eq. (23). For SNR = 100, the presence of noise has no measurable impact on determining the optimal phase angle for the range of frequencies considered, ensuring accurate results. For SNR = 50, the noise introduces some variability, resulting in a small range of optimal phase angles for which the mean absolute error is $\varepsilon \leq 10\%$ rather than two specific optimal values of φ_0. For SNR = 30, the error increases significantly, and this level of noise constrains the achievable accuracy of determining viscoelastic properties. However, even in this case, using the optimal phase angle reduces the noise level by approximately a factor of two.

Chirp signals with an initial phase of $-85°$ [Fig. 11(b)] produced spectra ($G'(\omega, t)$) with fewer artifacts in these early stages than chirps applied with a 0° phase angle [Fig. 11(a)]. To highlight this effect, we applied Gaussian process regression (GPR) [40] to estimate the posterior mean and uncertainty bounds. The GPR mean is shown as a dashed line in each plot, while the shaded envelope represents the associated uncertainty bound, offering a clear visual representation of the spectral confidence at each frequency [Figs. 11(a) and 11(b)]. As noted by Perego *et al.* [14] (see their Fig. 7), the average mutation time changes rapidly near the onset of initial curing with rapid cross-linking, between 150 and 350 s —a transition that coincides with significant changes in the viscoelastic response. During this interval, Figs. 11(a) and 11(b) show that using a non-zero initial phase shift ($\varphi_0 = -85°$) results in a smoother spectrum with smaller uncertainty bounds compared to a zero initial phase shift ($\varphi_0 = 0°$).

As the curing time progresses and the cross-link density increases, both values of the phase shift produced linear viscoelastic spectra of comparable quality. Thus, we did not include viscoelastic spectra from OWCh experiments beyond 568.5 s. This transition reflects the material's evolution into a more solid-like viscoelastic state. As gelation progresses toward the critical gel point, the material's longest relaxation time, as well as the zero-shear rate viscosity, diverges, and this "infinite relaxation mode" is subsequently taken up by the equilibrium (plateau) modulus as the material passes beyond the gel point. The remaining stress-relaxing modes then have a progressively shorter mean relaxation time as the cross-linked network strengthens and the response becomes increasingly elastic. From this viewpoint, the plateau modulus (which sets the scale for the stress output from the chirp) grows, while the relevant portion of the relaxation spectrum (which controls the rate of approach to the oscillatory steady state) contracts post-gelation [41]. As a result, in such solid-like systems, chirp signals with any initial phase shift generally perform well. Moreover, as the stress amplitude increases, the relative impact of noise from the signal's finite stress at the end, and from the asymmetry in the signal, remains negligible.

IV. IMPACT OF NOISE ON ACCURATE PREDICTION OF THE VISCOELASTIC MODULI

All of the previous numerical investigations evaluating the influence of the initial phase shift on the accurate estimation of viscoelastic moduli were conducted under ideal, noise-free conditions. Likewise, experimental tests on the micellar solution and high-strength acrylic adhesive 300 LSE transfer tape were carefully designed through selection of approapriate test geometry and strain amplitude to minimize external perturbations, ensuring that the effect of random noise (originating from the rheometer itself and the external environment) could be neglected. However, in practical scenarios, output stress signals often contain significant noise from the rheometer itself or external sources. This is particularly relevant when working with liquid-like materials (e.g., a UV-curable acrylate system at the beginning of the cross-linking process) or small test samples. Therefore, it is crucial to assess how rheometer noise characteristics affect the determination of the optimal phase shift.

To investigate this, we reanalyzed the error versus optimal phase angle curve from Fig. 6(b), this time incorporating the presence of noise in the stress output. The analytic stress was obtained using the CMM model with an input chirp signal corresponding to a large time-bandwidth product (TB $\approx$ 99). To simulate experimental conditions, we then added white Gaussian noise to the output stress signal using MATLAB®'s built-in "Additive white Gaussian noise" (awgn.m) function.

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Two sample stress output signals with signal-to-noise ratios (SNRs) of 100 and 30 are shown in Fig. 12(a). In Fig. 12(b), we present the error [Eq. (23)] versus optimal phase angle (φ_0) curves for three different SNR values: 30, 50, and 100. We observe that for SNR = 100, the conclusion that two optimal phase angles located 180° apart result in zero percent normalized mean absolute error (ε) remains valid. For SNR = 50, the presence of Gaussian noise introduces additional variability, leading to a narrow range of optimal phase angles with small (but non-zero) values of ε rather than two distinct φ_{opt} values. While achieving exactly zero percent error is no longer possible, the error remains relatively low, staying within $\leq 10\%$. However, for SNR = 30, even at the optimal phase angle, the mean absolute error (in the minor component of the complex modulus) is significant, reaching nearly 70%. At this noise level, the accuracy of viscoelastic property determination is severely compromised, but adjusting the phase angle still reduces the mean absolute error by a factor of two.

Notably increasing the final frequency, ω_2, amplifies the impact of noise, leading to pronounced distortions in higher frequency components of the material properties. This occurs because the input strain signal is tapered at both ends using a Tukey window to ensure a smooth transition, introduce periodicity, and reduce spectral leakage. A similar windowing effect is observed in the output stress too. High-frequency components of the instrument noise spectrum can overlap with the original viscoelastic spectrum, leading to the emergence of noise artifacts in chirp experiment data. More complex windowing functions and slew rates may be used to help offset such effects [42].

To summarize, the noise in viscoelastic material spectra determined from chirp experiments arises primarily from two factors. First, when T_{owc} is comparable to the material's inherent dominant relaxation timescale τ, the initial transient material response does not fully decay within the experimental test window, T_{owc}, leading to systematic errors in the measured values of the viscoelastic moduli. By tuning the input strain with an optimal phase shift, the accuracy of the resulting viscoelastic spectrum can be improved, provided that both the input strain and output stress signals remain free of noise. Second, when external random noise overlaps with the output signal, additional noise artifacts limit the spectral resolution. However, using a nonzero initial phase angle $\varphi_0 \simeq 90^\circ$ can still reduce the impact of this noise. Further optimizing the results obtained with chirp signals will require effective broadband noise reduction strategies.

V. CONCLUSION

In this study, we have examined the capabilities and limits of γ-OWCh-based rheometry for characterizing mutating materials with rapidly evolving viscoelastic properties. The technique utilizes an optimally windowed, strain-controlled chirp signal, $\gamma(t)$, to probe the material. Because the material's viscoelastic response (i.e., the components $G'(\omega)$ and $G''(\omega)$) are computed in the frequency domain, the time-domain signals must be free of distortions arising from transducer noise, baseline drift, or sharp discontinuities at the signal boundaries to ensure the accuracy of the predicted viscoelastic moduli.

Geri *et al.* [7] demonstrated that using a cosine-tapered window (Tukey window) minimizes spectral leakage and allows the use of chirp signals with a relatively short time-bandwidth product (TB < 100). However, we showed that in addition to considerations of the time-bandwidth product, TB, and the mutation number, *Mu*, the design of the chirp signal should also take account of the initial transient response arising from the material's dominant relaxation time, τ. In conventional methods, such as the DFS or multi-wave measurement techniques, the user can specify a waiting period (or delay time) so that the data are only collected after the material reaches a steady alternating state of stress. In contrast, chirp-based experiments repeatedly probe a material using short, highly structured signals and inherently probe the material during its unsteady transient response.

Our initial numerical investigations using a simple Maxwell model revealed that when the chirp duration is comparable to, or less than, the dominant relaxation time of the material, i.e., $T_{\text{owc}} \leq \tau$, the stress does not decay completely to zero at the end of the chirp. This incomplete relaxation contributes to an aperiodic stress response as well as resulting in a waveform that is asymmetric about the centerline. Such an aperiodic and asymmetric signal introduces noise into the frequency-domain representation. The underlying cause is that the transient part of the material response does not have sufficient time to relax within the experimental duration. To overcome these challenges, we have proposed treating the initial phase of the input OWCh signal, φ_0, as a variable tuning parameter rather than fixing it at zero. By introducing a finite φ_0, we can precisely control the temporal shift of the chirp signal. This phase adjustment realigns the input strain rate, $\dot{\gamma}(t')$, shifting the peaks and troughs so that its convolution with the relaxation modulus $G(t-t')$ in the Boltzmann superposition integral balances more symmetrically. In effect, the optimal phase shift, φ_{opt}, restores the temporal symmetry of the time- and frequency-dependent stress signal, as well as minimizing the residual stress at $t = T_{\text{owc}} + t_w$. This reduces frequency artifacts and unwanted noise in the Fourier domain.

To illustrate the impact of the initial phase shift, we generated contour plots, such as Fig. 6(a), showing the signal error [as quantified by the discrepancy in $G''(\omega)$—the subdominant component (classical Maxwell model)] for different combinations of $(\tau/T_{\text{owc}}, \varphi_0)$. Our findings can be summarized as follows:

- For any value of τ/T_{owc}, there exist two optimal phase angles (separated by a phase angle of 180° or π rad) that yield the smallest error.
- When $\tau/T_{\text{owc}} \ll 1$, the impact of the initial phase shift on the error is negligible. In this regime, all values of φ_0 result in high accuracy $G'(\omega)$, $G''(\omega)$.
- As τ/T_{owc} increases, the error curves exhibit two narrow regions of optimal φ_0 ($\approx \pm\pi/2$), which become nearly constant for large τ/T_{owc}.

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- The presence of random transducer noise in the measured output signal limits the reduction in the mean absolute data error that can be attained, but using a nonzero value of the initial phase angle (e.g., $\varphi_0 = \pi/2$) still improves the accuracy in determining the minor component of the complex modulus of the material.

These observations were further validated using computations with the fractional Maxwell model, which compactly captures the broad relaxation spectra of many viscoelastic materials. Finally, our experimental investigations on wormlike micellar solutions, a high-strength acrylic adhesive, and a UV-curable acrylate system confirmed the practical utility of using a nonzero initial phase. In all cases, when the material's dominant relaxation time was comparable to the chirp duration, adjusting the phase angle φ_0 of the input signal simply and significantly enhanced the accuracy of the subsequent viscoelastic measurements.

Although all numerical investigations and experimental analyses in this work were conducted using the γ-OWCh signal as input, selecting an optimal initial phase shift can also benefit controlled stress chirp (σ-OWCh) rheometry. In this case, it can again help realign the output strain rate $\dot{\gamma}(t)$ with the input stress $\sigma(t')$ (or more accurately the history of the stress loading rate, $\dot{\sigma}(t')$), potentially enhancing the quality of the resulting compliance spectrum determined in the Fourier domain.

To conclude, our results show that selecting an appropriate non-zero initial phase angle, φ_0 is an essential first step in chirp-based rheometry, as it directly improves temporal symmetry and reduces residual transients. In practice, phase values near $\varphi_0 \simeq \pm\pi/2$ rad provide a reliable starting point, which can subsequently be refined by inspecting the offset and $\pm$ symmetry of the stress signal at the end of the chirp. An even more experimentally robust protocol would pair such phase tuning with targeted denoising of the high-frequency region and, if notable aperiodicity or asymmetry persists, using a modified analysis based on an alternative transfer function, such as $\kappa^*(\omega)$ [Eq. (11)] which is insensitive to constant offsets.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

DATA AVAILABILITY

The data and code that support the findings of this study are available from the corresponding author upon reasonable request.

APPENDIX: OPTIMAL PHASE ANGLE FOR SHORT TB PRODUCT CHIRPS

To identify optimal phase angles for short time-bandwidth product chirps (TB < 10), numerical investigations were carried out using the classical Maxwell model [Eq. (12)]. For all simulations, the elastic modulus ($G_0 = 38$ Pa) was kept constant. The frequency range was set from $\omega_1 = 3$ rad/s to $\omega_2 = 30$ rad/s, and the chirp duration was defined as $T_{owc} = 2$ s (TB ≈ 8.6). The material relaxation time, τ was varied to achieve ratios ranging from $0.1 \leq \tau/T_{owc} \leq 2$. The resulting contour plot of the error, $\varepsilon(\varphi_0, \tau/T_{owc})$, calculated using Eq. (23) is shown in Fig. 13(a). To highlight regions of high accuracy, we marked areas where the mean absolute error is less than 5% with dashed lines.

Comparison of the viscoelastic moduli, $G'(\omega)$, $G''(\omega)$, obtained for phase shifts $\varphi_0 = 0°$, $\varphi_{opt} = -92.7°$ with the

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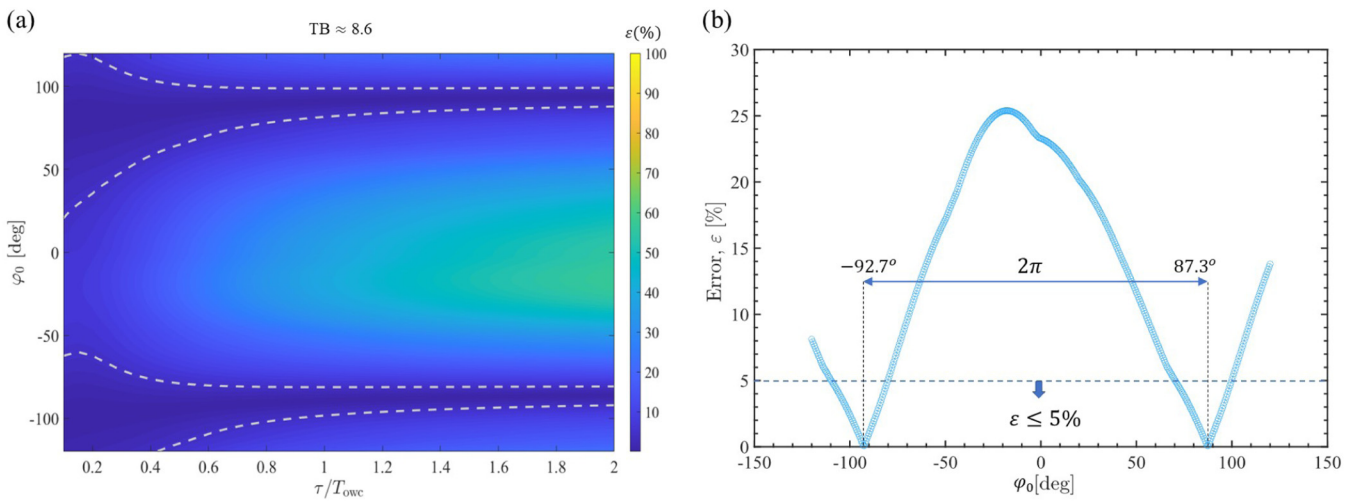


FIG. 13. (a) Contour plot of the error, $\varepsilon(\varphi_0, \tau/T_{owc})$, computed using Eq. (23) for a short time-bandwidth product chirp signal (TB ≈ 8.6). The chirp parameters used are $\omega_1 = 3$ rad/s, $\omega_2 = 30$ rad/s, $T_{owc} = 2$ s, and $r = 10\%$. For the Maxwell model parameters, $G_0 = 38$ Pa is kept fixed and τ varies from 0.2 to 4 s, resulting in a ratio ranging from $0.1 \leq \tau/T_{owc} \leq 2$. The dashed lines indicate regions where the error $\varepsilon \leq 5\%$. (b) 2D plot of error, ε [Eq. (23)], for a specific case of $\tau/T_{owc} = 1$. The dashed line highlights the range of phase angles where $\varepsilon \leq 5\%$. For the specified chirp parameters, two locally optimal phase angles ($-92.7°$ and $87.3°$) separated by $180°$ result in a zero percent mean absolute error ($\varepsilon = 0\%$).

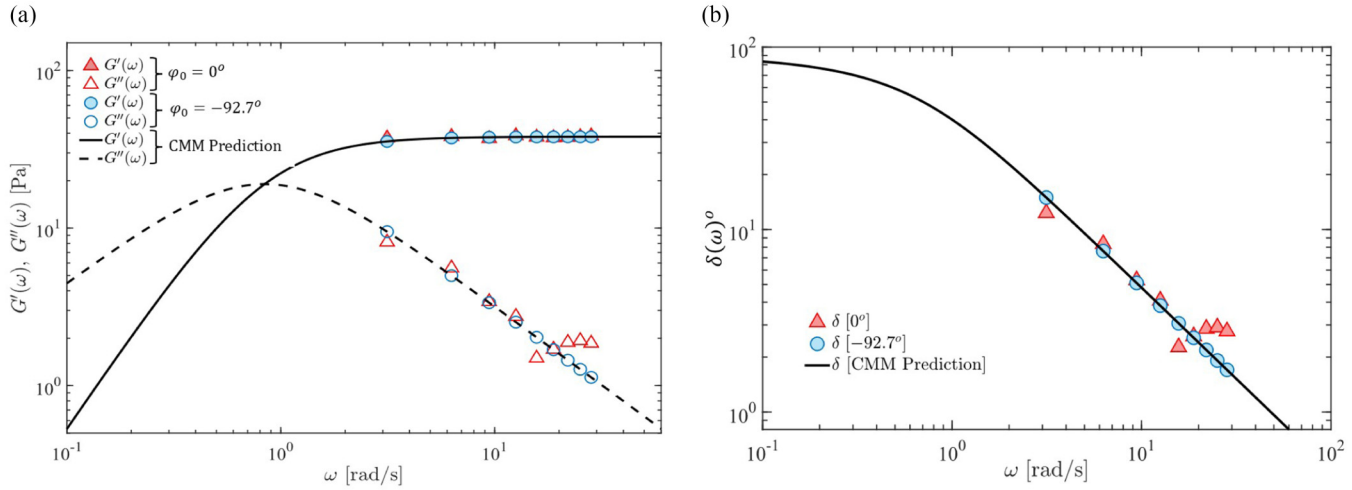


FIG. 14. Data presented are obtained by numerically solving Eq. (12) for an input γ-OWCh waveform with $G_0 = 38$ Pa and $\tau = 1$ s. (a) Comparison of viscoelastic moduli, $G'(\omega)$ and $G''(\omega)$, obtained from windowed chirp signals with different initial phase shifts. Data for $\varphi_0 = 0°$ are represented as filled triangles ($G'(\omega)$) and open triangles ($G''(\omega)$), while data for $\varphi_0 = -92.7°$ are shown as filled circles ($G'(\omega)$) and hollow circles, ($G''(\omega)$). Reference predictions from the CMM for $G'(\omega)$ and $G''(\omega)$ are shown as black solid and black dashed lines, respectively. (b) Comparison of the computed phase angle, $\delta(\omega) = \tan^{-1}(G''(\omega)/G'(\omega))$ for $\varphi_0 = 0°$ and $\varphi_0 = -92.7°$. The same symbols and colors as in (a) are used.

ground truth (from the CMM predictions), is presented in Fig. 14(a). Similarly, in Fig. 14(b), we compare the phase angle, $\delta(\omega)$, obtained for different initial phase shifts with the ground truth. The results demonstrate excellent agreement between the ground truth and the values of the viscoelastic moduli obtained with an optimally windowed chirp with optimal phase shift, as well as between the phase angle computed with the optimal phase shift and the ground truth.

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